

Enhancing the performance of class G cement for subsurface gas storage and well completion: Synergistic impact of North Dakota's fly ash and eggshell powder

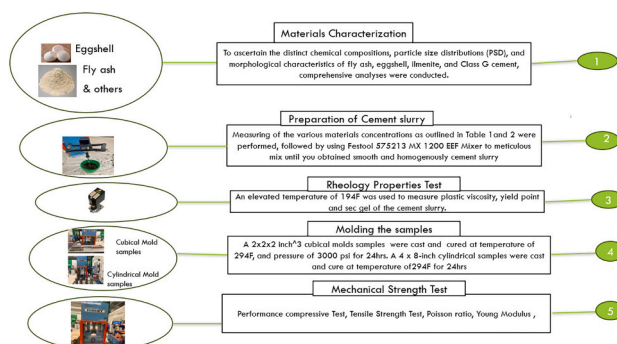
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HIGHLIGHTS

- Study validates synergy of fly ash & eggshell for enhancing cement.
- The additive improves elastic properties, enhancing durability.
- Cement matrix porosity & permeability were greatly reduced.
- Enhances cement compressive strength, gel strength, yield stress & viscosity.
- 2% of the blend improves cement geo-mechanical integrity.

GRAPHICAL ABSTRACT



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ABSTRACT

Sustainable well completions and integrity become critical in response to the rapidly increasing growth in oil and gas production brought about by unconventional drilling. Ensuring durable zonal isolation behind casing is essential to prevent leakage and safeguard subsurface inter-zone communications, critical for overall well integrity. The longevity and safety of underground gas storage hinge on the performance of the cement employed in well completions. Class G cement represents more desirable alternatives to conventional Portland cements, but unreconciled deficiencies pose risks to its reliable application, particularly in highly volatile hydrocarbon well completions and underground gas storage. This study delves into the synergistic effects of fly ash and eggshell powder (FAESP) on Class G cement performance, aiming to enhance its mechanical strength, durability, and resistance to the extreme downhole environments encountered in oil and gas fields wells. Fly ash, a well-recognized pozzolanic material, and eggshell powder, a bio-based additive, were strategically chosen for their potential to improve cement properties and contribute to sustainable well completions for underground gas storage. The materials, including Class G cement powder, fly ash, eggshell powder and ilmenite were characterized using XRF analysis, particle size distribution and SEM-EDX analysis. A comprehensive set of tests were conducted to evaluate the effects of varying concentrations of fly ash and eggshell powder (FAESP) ranging from 0.25% to 4.0% by weight of cement (BWOC) on class G cement and benchmarked against plain class G cement (Base cement). The study encompassed a series of tests which include rheological parameters; compressive and

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tensile strengths; elastic attributes (Young's modulus and Poisson's ratio); and petrophysical characteristics such as porosity and permeability assessment, simulating the harsh reservoir conditions. The results indicated that the optimum fly ash (FA) and eggshell powder (ESP) concentration was 2% by weight of cement (BWOC), based on wellbore-relevant performance tests. This concentration significantly improved yield stress and gel strength while reducing porosity and permeability, addressing gas migration concerns. It also enhanced compressive strength and imparted resilience, as evidenced by a reduction in Young's Modulus. This first-of-its-kind study validates the potential for sustainable and cost-effective advancement of ilmenite-cement properties by harnessing the synergistic effects of combining fly ash and eggshell powder. The findings demonstrate the feasibility of ensuring durable zonal isolation over the well life cycle.

1. Introduction

Mitigating subsurface gas migration and well integrity issue while preserving energy resources from posing environmental risks has emerged as a formidable challenge for scientists and researchers. To address this challenge, there is a concerted effort among researchers and engineers to explore alternative materials that can substitute cementitious components while simultaneously minimizing gas leakage and enhancing wellbore integrity in oil and gas wells [1]. Among the array of industrial waste products, fly ash and eggshell powder have surfaced as promising candidates for cement replacement due to their cementitious properties and potential to alleviate disposal issues associated with these abundant materials [2]. Composition of cement slurry are critical during the drilling and completion processes of underground storage wells. An effective cementing job is essential for achieving reliable zonal isolation and safeguarding the casing integrity [3]. The cement behind the casing in oil or underground gases storage's wells is subjected to various forces, including casing load, formation pressure, and hydrostatic head [4]. Consequently, investigating the properties of oil or gases storage well cement under these conditions is crucial for ensuring the integrity and performance of the well [5]. Inadequate cement design can not only lead to reduced oil production but also have severe environmental consequences, particularly in marine habitats [6]. This was evident in the Deepwater Horizon accident, where cementing failure at a depth of 18,000 ft. resulted in catastrophic environmental [7]. Therefore, designing an appropriate cement slurry and implementing efficient cementing operations are of utmost importance in the oil or underground gases storage wells operations [8], however, improper design of cementing

operation will lead to storage gas or oil leakage along wells as illustrated in Fig. 1. Achieving zonal isolation hinges upon cement formulation performance in providing durable, impermeable barriers to fluid migration behind casing [9,10]. Class G cements have risen as more sustainable alternatives to conventional Portland oil well cements, with 95% of industries worldwide utilizing this class of cement, yet still suffer deficiencies including brittleness, shrinkage cracking, and carbon dioxide susceptibility that can compromise sealing integrity ([11]; Mangadlao et al., 2015). The main constituents of Portland cement include lime (CaO), silica (SiO_2), alumina (Al_2O_3), iron oxide (Fe_2O_3), and gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) [12]. These raw materials undergo a series of complex hydration reactions, resulting in the formation of four principal compounds that constitute Portland cement [13]. These compounds are tricalcium aluminate ($\text{Ca}_3\text{Al}_2\text{O}_6$ or C_3A), tricalcium silicate (Ca_3SiO_5 or C_3S), tetracalcium aluminoferrite ($\text{Ca}_4\text{Al}_2\text{Fe}_2\text{O}_{10}$ or C_4AF), and dicalcium silicate (Ca_2SiO_4 or C_2S) [14]. The hydration process of these compounds plays a crucial role in the development of the cement's strength and other essential properties required for its application in drilling operations [15]; [16]. Depending on the existing conditions within the wellbore and the specific goals of the cementing operation, various additives are blended into the cement slurry formulation to ensure the creation of an effective cement bond [17]. With advancements in technology, the oil and gas industry are actively seeking to incorporate advanced materials and equipment in the development of more effective cement slurries for wellbores [18]. These efforts aim to enhance hydrocarbon extraction while minimizing losses and ensuring environmental protection caused by underground gas migration through the cement sheath [19]. By leveraging innovative solutions and best

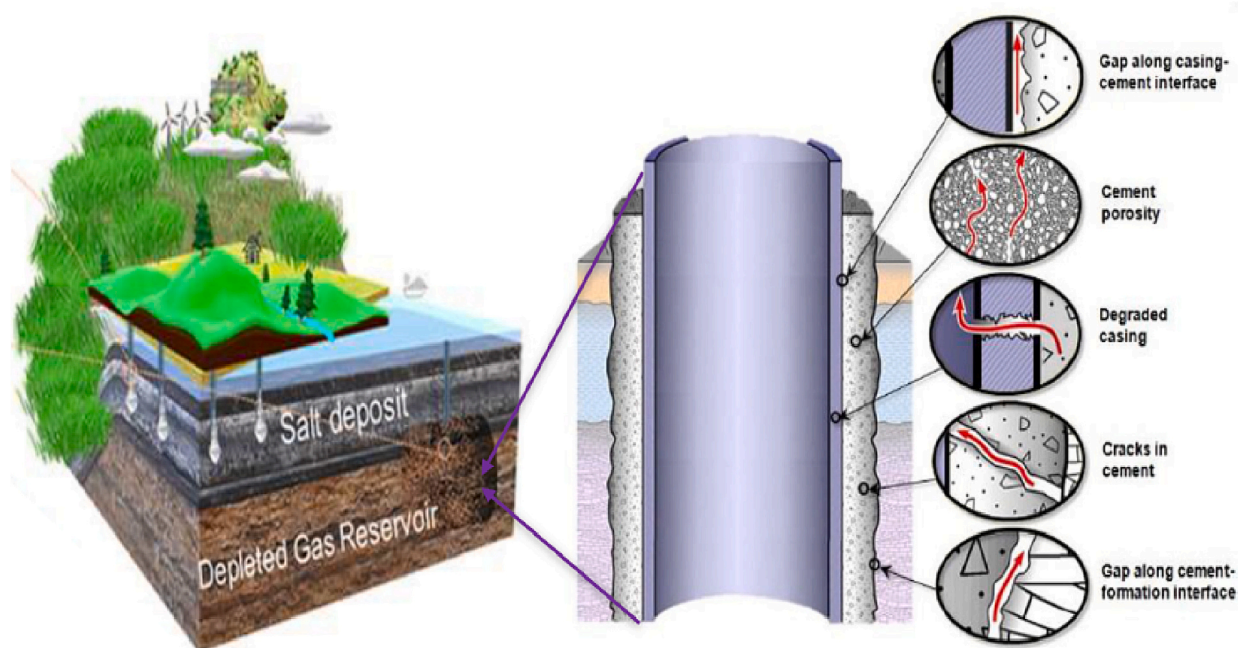


Fig. 1. Improper design of cementing operation led to storage gas or oil leakage along wells.

practices, the industry strives to optimize cementing operations and mitigate potential risks associated with improper cementing design [20]. The quest for enhancing the performance of cement used in oil and gas well completions has led to the exploration of various additives that can improve the mechanical strength, durability, and resistance to harsh downhole environments [21,22]. Class G cement, commonly used in subsurface oil and gas exploitation and storage applications, must withstand extreme conditions, including high temperatures, pressures, and mechanical stresses [23]. There is urgent needs for the energy and petroleum industry to optimal the G class cement that can tolerate the hostile gas storage environment and sustaining its properties over the period of gas storage. In this study, the incorporation of industrial by-products such as fly ash and eggshell powder has garnered significant attention due to their potential to improve cement properties and contribute to sustainable practices [24]. Fly ash, a by-product of coal combustion, has been extensively studied for its pozzolanic properties, which can enhance the mechanical and durability characteristics of cementitious materials ([25]; [26]; [27]). Also, according to Amin et al., [28] and Fadairo and Oni [29], industrial byproducts like fly ash and eggshell powder have shown aptitude for enhancing cement properties through pozzolanic interactions and microstructural modifications during hydration binding processes. Fly ash contains abundant silica and alumina species to augment formation of calcium-silicate-hydrate (C-S-H) gels that improve strength and permeability [30]. (See Table 1.)

Fly ash is a by-product generated from the combustion of pulverized coal in powder stations or other coal-burning facilities for electricity or energy production [40]. It is the most abundant form of coal ash, accounting for 75–85% of the total ash produced [41]. Fly ash is an amorphous, fine, powdery material that predominantly consists of spherical particles (solid or hollow) and is typically tan, grey, or black in color which can be readily synthesized into ultra-fine particles owing to their exceptionally low density [42,43]. The color of the fly ash is generally indicative of its carbon content, with lighter colors corresponding to lower carbon levels [44]. The particle size of fly ash ranges from 0.5 μm to 100 μm [45]. Numerous studies have been conducted to investigate the beneficial applications of coal fly ash [46]. These applications include its use in agriculture, catalysis, and as a buffering material [47]. Therefore, it is imperative to explore the increased utilization of an enhanced form of fly ash to mitigate the consumption of natural resources in cement manufacturing [48]. Additionally, coal fly ash has shown promising advantages as a filler material for polymers [49]. The incorporation of coal fly ash into a polymer matrix has been associated with improvements in the processability of the polymer and enhancements in properties such as tensile strength and heat-distortion temperature [50]. Previous study has also demonstrated that the compressive strength of concrete increases as the percentage of fly ash and rice husk ash increases, up to a certain replacement level of cement [51]. Joel 2020 reported that the optimum replacement level for achieving the highest compressive strength was found to be 22.5% fly ash and 7.5% rice husk ash in concrete with different mixing ratios. The addition of fly ash to cement slurry can improve the rheological properties, making it more workable and easier to pump, although it may reduce the compressive strength if used in higher percentages ([52]; [53]; [39]). The combination of fly ash and eggshell powder in Class G cement could potentially offer a synergistic effect, improving the overall performance of the cement in well completions. The lightweight nature of fly ash-based cement systems, combined with the intrinsic thixotropic behavior, can address issues such as loss of circulation and provide high early-age mechanical properties [54]. The use of these additives aligns with the industry's move towards more sustainable and cost-effective solutions for well construction and maintenance.

Eggshell powder has emerged as a promising additive for cement concrete and oil and gas cementing applications due to its unique composition and properties. Eggshells, which are often considered as waste material from the food industry, are rich in calcium carbonate (CaCO_3), constituting about 95% of their composition [55]. This high

Table 1

Some applications of fly ash and eggshell powder in cementing.

Study	Materials	Application	Conclusions/Findings
Singh and Choudhary [31]	Microconcrete, Fly ash	Cement replacement	Fly ash enhanced compressive strength, ductility, and durability.
Khan and Lynsdale [32]	Class G cement, Fly ash	Oil well cement	Fly ash increased thickening time and lowered fluid loss.
Cheah and Ramli [33]	Mortar, Eggshell powder	Cement replacement	Eggshell powder reduced water permeability and porosity.
Yuan et al. [34]	Cement paste, Eggshell powder	Cement substitution	Eggshell powder refined pore structure and microstructure
Gesoğlu et al. [35]	Mortar, Fly ash	Cement replacement	Fly ash delayed hydration and lowered early strength but improved later strength.
Zareei et al. [36]	Mortar, Eggshell powder	Cement substitution	Eggshell powder increased compressive strength and lowered chloride permeability.
Kandasamy and Shehata [37]	Concrete, Eggshell powder	Cement replacement	Eggshell powder enhanced mechanical properties and microstructure at 15% replacement.
Gesoğlu et al. [35]	Mortar, Fly ash	Cement replacement	Fly ash delayed hydration and lowered early strength but improved later strength
Jhatial et al. [38]	Portland Cement, Eggshell Powder	Road construction	Eggshell powder enhances the workability of cement mixtures.
Chuan et al. (2022)	Geopolymer, Eggshell powder	Alkali activator modification	Eggshell powder-modified alkali activator improved geopolymer strength
Guo et al. [39]	Coal-Fly Ash, Cement	Cement-based grout	Fly ash improves rheological properties, reduces hydration exothermic rate, and increases porosity, leading to decreased compressive strength. Optimal fly ash content is 20–30% for better rheological properties and strength.

calcium content makes eggshell powder a potential substitute for conventional limestone fillers in cement-based materials [56]. Eggshell powder provides extra calcium to additionally support C-S-H development [57]. Further, the fine particle sizes and morphologies beneficially reduce porosity while accelerating hydration reactions. However, the specific influences of these supplementary materials on Class G oil well cements remain sparsely explored. Eggshell powder has been investigated for its potential use in various construction materials [58]. While the specific effects of eggshell powder on Class G cement for oil and gas well completions are less documented, research on its use in asphalt materials has shown that it can increase the consistency, hardness, and thermal stability [59]. These properties suggest that eggshell powder could similarly influence the high-temperature performance of cement used in well completions. However, further research is needed to fully understand the interaction mechanisms and the impact of eggshell powder on the mechanical and durability properties of Class G cement in the context of oil and gas wells [59]. Additionally, the visco-plastic flow behavior of cement slurries in squeeze cementing operations, as discussed in the context of repairing leaking wells, could be influenced by the addition of eggshell powder, potentially improving the flow characteristics and effectiveness of cementing operations [60]. Several studies have investigated the use of eggshell powder as a partial

replacement for cement in concrete. [61] found that replacing cement with up to 5% eggshell powder resulted in improved compressive strength and reduced water absorption of concrete. Similarly, Ngayakamo et al. [62] reported that the incorporation of eggshell powder enhanced the mechanical properties and durability of concrete, attributing these improvements to the fine particle size and high calcium content of eggshell powder.

2. Methodology

2.1. Material used

The research conducted in this study utilized several materials including Class G cement, defoamer, retarder, water, fluid loss additive, eggshell powder, North Dakota's fly ash, silica solution was sourced from Nouryon Pulp and Performance Chemicals Inc. and ilmenite was supplied from Trinity Ceramic Supply, Inc. in the United States. API Class G cement utilized in this study was sourced from Dyckerhoff Company, is highly recommended for various construction of casing and wellbore lining for oil and gas wells applications. While the fly ash was sourced from North Dakota coal powder plant station, the chemical composition of the fly ash includes Sulfur trioxide (SO_3), alumina oxide (Al_2O_3), silica dioxide (SiO_2), sodium oxide (Na_2O), iron oxide (Fe_2O_3), Phosphorus pentoxide (P_2O_5), calcium oxide (CaO), magnesium oxide (MgO) and titanium dioxide (TiO_2). Chemical analysis of the fly ash was conducted previously by [63] at University North Dakota Lab.

2.1.1. Eggshells powder

Eggshells, predominantly composed of Calcium Carbonate (CaCO_3), are rich in calcium content, which contributes to the enhancement of concrete strength. The eggshells were collected from the university restaurants and apartments. To prepare the eggshells for use in this research, they were initially thoroughly rinsed with water from a tap, during which the egg membranes were meticulously removed, it was then oven-dried for a duration of 1 h at a temperature of 90°C . Subsequently, the dried eggshells were pulverized into fine particles seen in Fig. 2 and were then passed through sieves with a mesh size of $20\ \mu\text{m}$ to ensure uniform particle size, adhering to the specifications set forth by ASTM C618–12.

2.2. Characterization and synthesis methods

The elemental compositions of Dyckerhoff Class G cement powder, eggshell and fly ash and ilmenite were assessed through X-ray Fluorescent (XRF) analyzers. The morphology and surface characteristics of the samples were examined using a scanning electron microscope (SEM) in Fig. 3. Additionally, the chemical elemental composition of the fly ash and eggshell powder (FAESP) and base sample (BS) were analyzed through Energy Dispersive X-ray Spectrometry (EDS) analysis. The

particle size distribution of the cement powder, fly ash, eggshell powder and ilmenite used in the cement slurry preparation was determined using the hydrometer analysis in the Lab.

2.3. Preparation of cement slurry

The preparation of the cement slurry adhered to the testing guidelines outlined in API RP 10 B. as recommended by the American Petroleum Institute. The composition details of each cement mixture are presented in Tables 2 and 3. Initially, all dry powders including cement, fly ash, eggshell, and ilmenite were accurately weighed and homogeneously mixed before the adding up with the of additives. Maintaining a water-to-cement ratio (w/c) of 0.44, water was introduced into a sturdy mixing container. The Festool 575,213 MX 1200 EEF mixer, operating at 1000 rpm, received all liquid additives (defoamer, silica solution,) first, followed by the incorporation of solid additives (fluid loss, retarder, and dispersant) which were initially mixed thoroughly with small amount of water to ensure a homogenous mixture before introduced of the dry powders. Subsequently, the cement/dry powder blend was gradually introduced uniformly into the sturdy mixing container containing fluid additives while the Festool mixer was running at 1000 rpm. It is generally recommended to add the dry powder/cement slowly to avoid clumping. Operating the Festool 575,213 MX 1200 EEF mixer at 100 rpm for 2 min facilitated the preparation of smooth and homogenous cement slurry. Fig. 4 illustrates the materials and equipment used in this research providing a brief overview of the workflow for preparing cement slurry and conducting rheology properties test and mechanical tests in the study.

To optimize the mixing ratio of fly ash and the eggshell used in the formulation of the cement slurry, a comprehensive study was conducted involving the formulation of various cement slurry with varying ratio amount of fly ash and eggshell powder. The base formulation adhered to API standards and utilized Class G oil well cement. The aim was to assess the individual and combined effects of these additives on the cement slurry's properties. The initial formulation, detailed in Tables 2, consisted of four distinct cement slurry concentrations. The control sample (S1) contained only the base cement slurry. The subsequent three samples incorporated fly ash and eggshell powder as partial replacements for cement by weight of cement (BWOC) at 25%, 50%, and 75%, as outlined in Tables 2. To evaluate the synergistic impact of combining eggshell powder and fly ash, additional cement slurry mixtures were prepared. These mixtures included combinations of eggshell powder (ESP) and fly ash (FA) in varying ratios, as specified in Table 2. Each formulation was meticulously mixed and tested to determine its influence on the cement slurry's performance characteristics, with the overarching goal of optimizing the cement design for enhanced oil and gas well completions. After identifying the most effective formulation with a proportion ratio of 75%/25% FAESP, this optimized ratio was utilized to prepare the final cement slurry series. The series comprised different

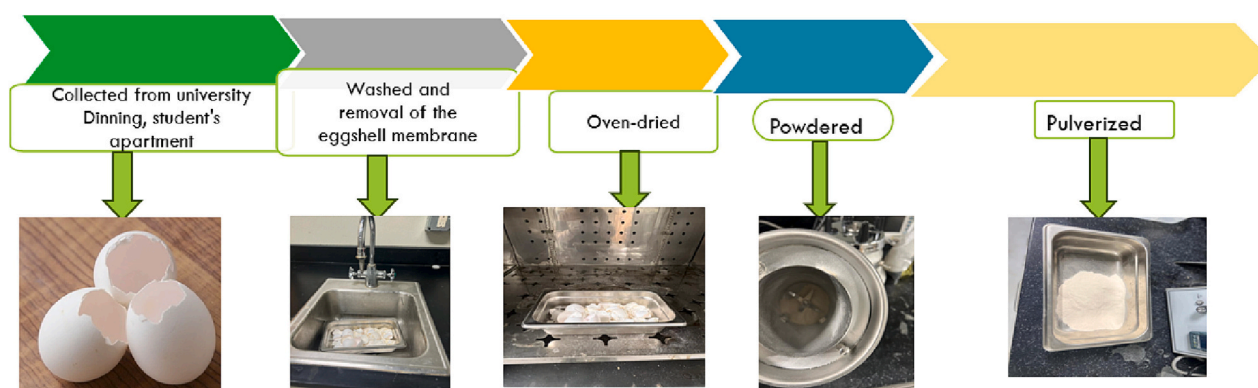


Fig. 2. The process of preparing pulverized eggshell powder.

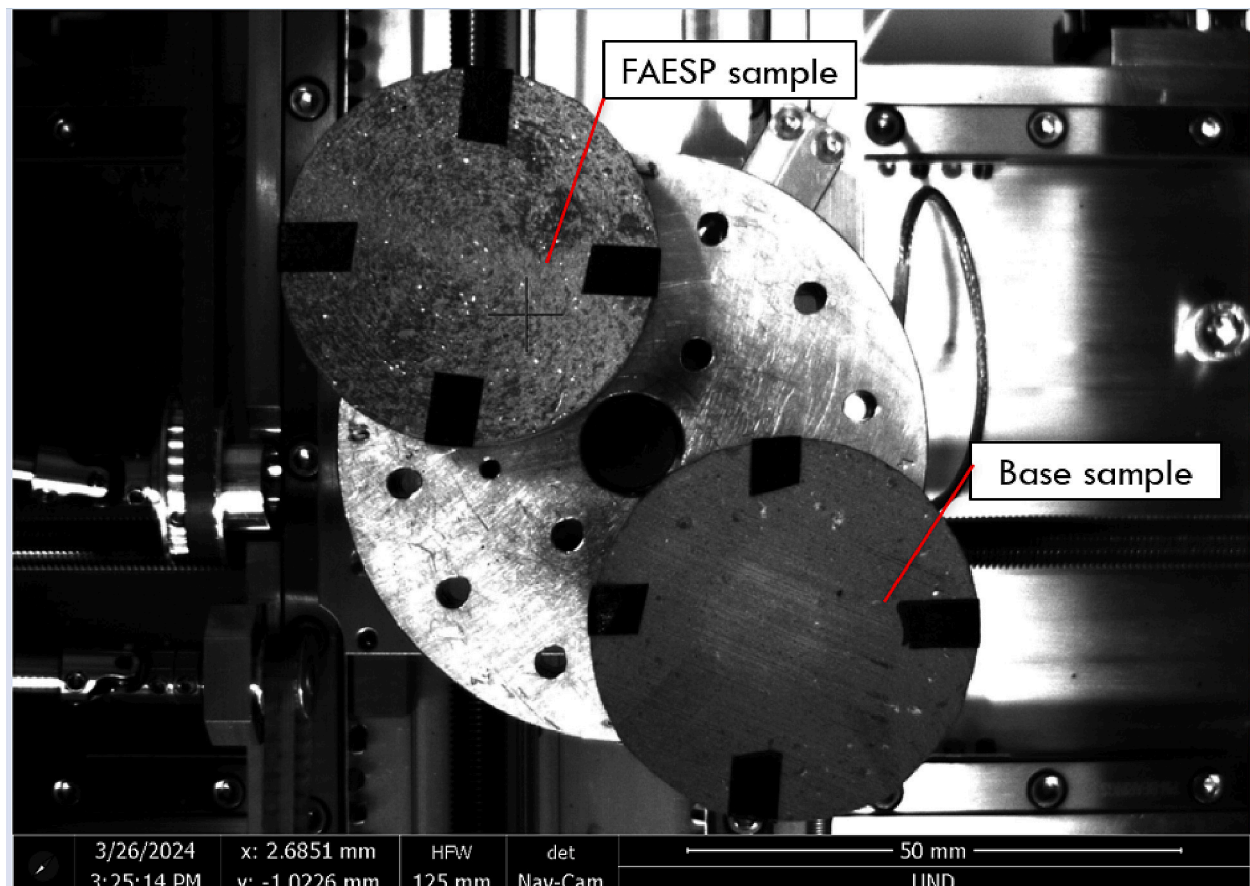


Fig. 3. Show the analysis of FAESP sample and Base sample with the using of SEM and EDX.

Table 2

Optimize cement slurry formulations.

Components	Base Cement Slurry	FA/ESP (75/25)	FA/ESP (25/75)	FA/ESP (50/50)	FA/ESP (75/0)
Cement (% BWOC)	100	100	100	100	100
Ilmenite (%)	36.3	36.3	36.3	36.3	36.3
Silica solution	35	35	35	35	35
Water (% BWOC)	44	44	44	44	44
Defoamer (% BWOC)	7	7	7	7	7
Dispersant (% BWOC)	0.25	0.25	0.25	0.25	0.25
Fluid Loss ((% BWOC)	0.5	0.5	0.5	0.5	0.5
Retarder (% BWOC)	1.5	1.5	1.5	1.5	1.5
Eggshell Powder (% BWOC)	0	75	25	50	0
Fly Ash (% BWOC)	0	25	75	50	75

concentrations of the combined fly ash and eggshell powder additives, ranging from 0% to 4% BWOC, as outlined in Table 3. Each increment within this range was systematically integrated into the cement slurry to assess its impact on the final properties. The prepared slurries underwent rigorous testing to validate the anticipated performance enhancements resulting from the partial inclusion of the optimal ratio of fly ash to eggshell. These tests' outcomes aimed to support the potential application of the optimized cement slurry in oil and gas well completions, ensuring improved well integrity and zonal isolation.

The median particle size (D50) and the particle size at which 90% of

Table 3

Final formulation cement slurry compositions.

Components	Base Cement Slurry	S 1	S 2	S 3	S 4	S 5
Cement (% BWOC)	100	100	100	100	100	100
Ilmenite (%)	36.3	36.3	36.3	36.3	36.3	36.3
Silica solution	35	35	35	35	35	35
Water (% BWOC)	44	44	44	44	44	44
Defoamer (% BWOC)	7	7	7	7	7	7
Dispersant (% BWOC)	0.25	0.25	0.25	0.25	0.25	0.25
Fluid Loss ((% BWOC)	0.5	0.5	0.5	0.5	0.5	0.5
Retarder (% BWOC)	1.5	1.5	1.5	1.5	1.5	1.5
Fly Ash and Eggshell Powder (FAESP) (% BWOC)	0.0	0.5	2.5	1.0	2.0	4.0

the sample is smaller (D90) were determined for Class G cement, Eggshell powder, ilmenite, and fly ash to evaluate their impact on the cement slurry for oil and gas well casing operations. The results, depicted in Fig. 4, indicated that the particle size distribution for Class G cement, eggshell powder, ilmenite, and fly ash spanned from 20 μm to 60 μm , 38 μm to 83 μm , 17 μm to 22 μm , and 45 μm to 80 μm respectively, as evidenced by the respective D50 and D90 analysis.

Furthermore, SEM imaging, as depicted in Figs. 4, characterized the morphology of individual eggshell, ilmenite and fly ash particles. The spherical morphology of fly ash particles enables efficient packing and sealing in the cement paste matrix, as indicated by previous studies [64,25]. Ilmenite particles exhibit a range of shapes from angular to rounded [63]. Eggshell powder possesses an irregular, flaky, and porous structure that creates sites for enhanced mechanical interlocking

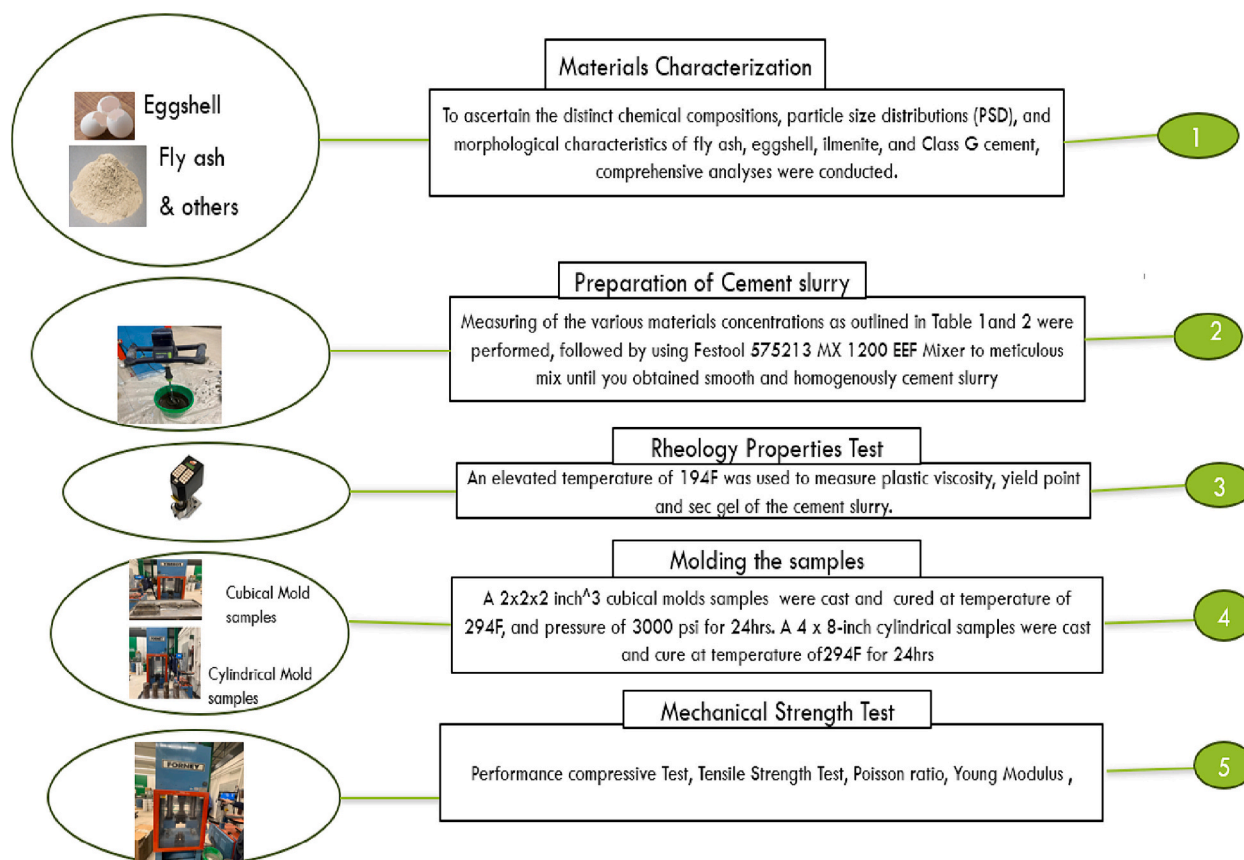


Fig. 4. Workflow for materials characterization, cement slurry preparation, rheology test and mechanical strength test.

[65,66,67]. X-ray fluorescence (XRF) spectroscopy provided quantitative insights into the chemical composition of fly ash from lignite sources, eggshell powder, and ilmenite. The primary elemental composition of ilmenite includes iron (III) oxide (Fe_2O_3), titanium oxide (TiO_2), magnesium oxide (MgO), silicon oxide (SiO_2), and aluminum oxide (Al_2O_3), with trace elements such as phosphorus pentoxide (P_2O_5), calcium oxide (CaO), and sulfur trioxide (SO_3). The major oxides in fly ash consist of silicon oxide, calcium oxide, sodium oxide, iron (III) oxide, aluminum oxide, sulfur trioxide, and magnesium oxide, with minor potassium oxide. X-ray fluorescence (XRF) spectroscopy provided quantitative insights into the chemical composition of fly ash from lignite sources, eggshell powder, and ilmenite. The primary elemental composition of ilmenite includes iron (III) oxide (Fe_2O_3), titanium oxide (TiO_2), magnesium oxide (MgO), silicon oxide (SiO_2), and aluminum oxide (Al_2O_3), with trace elements such as phosphorus pentoxide (P_2O_5), calcium oxide (CaO), and sulfur trioxide (SO_3). The major oxides in fly ash consist of silicon oxide, calcium oxide, sodium oxide, iron (III) oxide, aluminum oxide, sulfur trioxide, and magnesium oxide, with minor potassium oxide as illustrated in Tables 4 to 6 and Figs. 5 to 7. The chemical composition of eggshell powder includes calcium oxide (CaO), with minor amounts of silicon oxide (SiO_2), magnesium oxide (MgO), aluminum oxide (Al_2O_3), phosphorus pentoxide (P_2O_5), and sulfur

trioxide (SO_3), as determined by Bashir et al. [68] and Oni et al. [25]. These detailed analyses contribute to a comprehensive understanding of the physical and chemical characteristics of these additives, providing valuable information for optimizing cement slurry in oil and gas well completions.

2.4. Optimization of FAESP ratio of concentration

In this study, we first establish the optimal FAESP ratio of concentration that was used for our final cement slurry formulation. This was done by varying the concentration ratios of FA/ESP including FA/ESP (75%/25%), FA/ESP (25%/75%), FA/ESP (50%/50%) and FA/ESP (75%/0%) and this is a critical phase of the study. Comprehensive tests were carried out on rheological properties, mechanical properties and petrophysical properties to identify the most effective concentration ratio for the final formulation. The rheological properties assessments involved the measurements of plastic viscosity, yield point, and 10-s gel strength, which are pivotal for predicting the slurry's behavior under dynamic conditions. Also, the mechanical properties analysis included the determination of compressive strength, tensile strength, Poisson's ratio, and Young's modulus providing a holistic understanding of the wellbore's structural integrity and elastic characteristics for underground gases storage (Hydrogen and CO_2). The empirical data derived from these comprehensive tests indicate that the FA/ESP (75%/25%) ratio emerged as the optimal concentration for the final cement slurry formulation. This optimal ratio demonstrated a harmonious balance between workability and mechanical robustness, ensuring enhanced performance characteristics of the cementitious material.

2.5. Measurement of rheological properties

The rheological properties of oil well cement slurries are typically

Table 4
Testing the fundamental characteristics of materials.

Materials	Specific gravity	Median particle size D50 (μm)	D90 μm
Class G cement	3.12	20	60
Eggshell powder (ESP)	2.5	45	80
Fly ash (FA)	2.38	38	83
Ilmenite	4.29	17	22

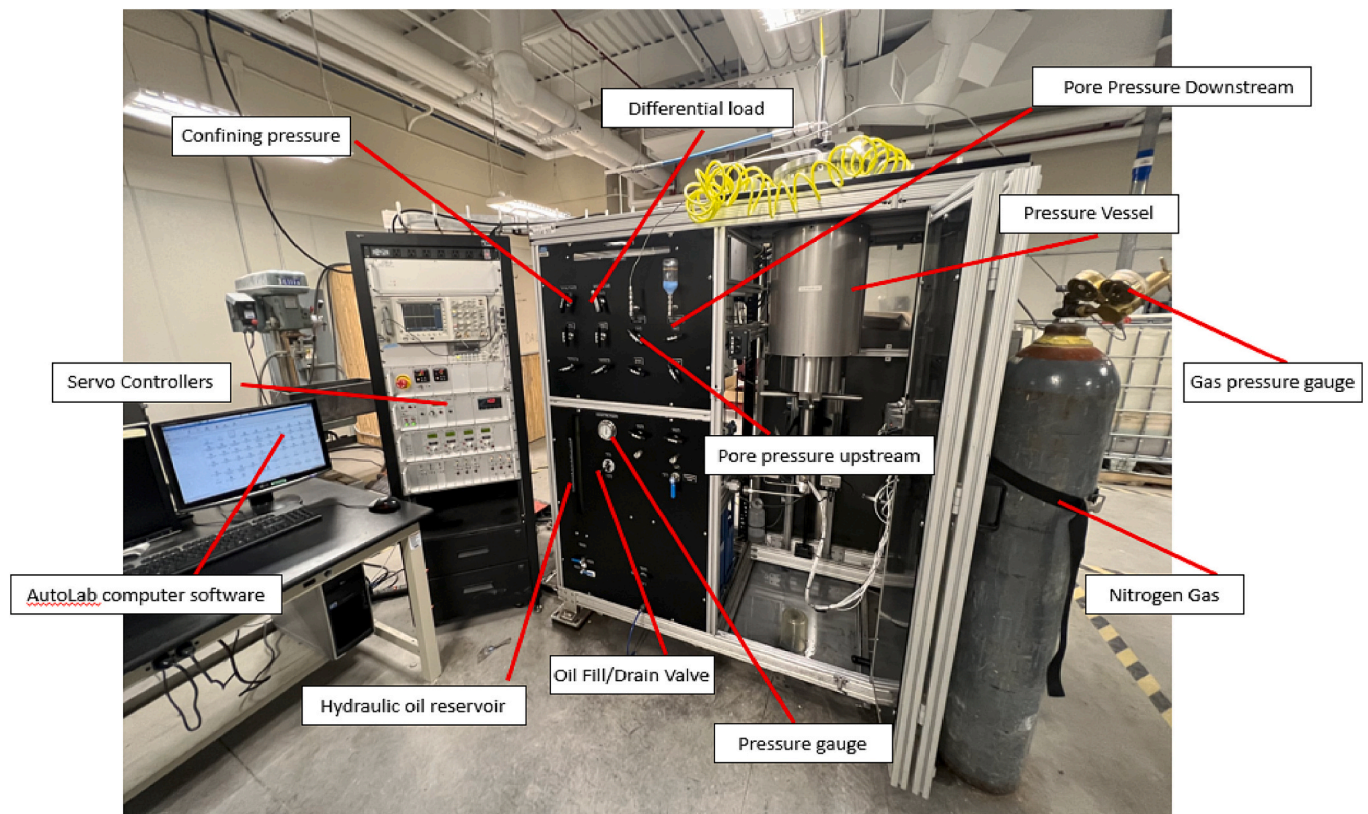


Fig. 5. Auto Lab 1500 multi-purpose test system.

measured using rotational viscometers at different temperatures. These measurements include apparent viscosity, yield stress, and plastic viscosity at various shear rates, which provide insights into the quality and physical characteristics of the slurry [69]. The significance of these properties lies in their ability to predict the performance of the cement slurries in oil well applications, such as ensuring proper placement, setting, and long-term stability of the cement sheath. Chemical admixtures, such as superplasticizers, have been used to modify the rheological properties of oil well cement slurries. These admixtures influence the fluidity and workability of the cement slurries at different temperatures and dosages, and their effects on the rheology of the slurry depend on the type and dosage of the admixture used (Tao et al., 2020). These properties were examined at a heightened temperature of 194 °F using an M-900 viscometer. The slurry was first conditioned, then transferred into the viscometer's cup for testing. Key parameters such as gel strength, yield point, and plastic viscosity were determined by recording the dial readings at multiple shear rates (600, 300, 100, 6, and 3 RPM), both increasing and decreasing. These readings were used to compute the shear stress at 300 and 600 RPM, which provided insights into the gel strength, yield point, and plastic viscosity of the slurry samples.

The rheological properties were quantified employing the following equations:

$$\text{Apparent Viscosity (AV)} = 600/2$$

$$\text{Plastic Viscosity (PV)} = \Phi 600 - \Phi 300$$

$$\text{Yield Point (YP)} = \Phi 300 - \text{PV}$$

2.6. Measurement of elastic properties

The structural performance of cementitious materials is heavily influenced by crucial mechanical properties such as the Young's

modulus (YM) and Poisson's ratio (PR). This study focused on the elastic properties of cement samples with varying fly ash and eggshell powder (FA/ESP) concentration, to assess the mechanical properties of the samples, a Forney 920 Series Universal Testing Machine with Automatic (VFD) Variable Frequency Drive controls was employed. Initial assessments involved cylindrical cement samples measuring 4 in. in length and 2 in. in diameter. These samples, with proportion ratios of ESP/FA (25% + 75%), FA/ESP (75% + 25%), FA/ESP (50% + 50%), and FA-S4 (75%), were subjected to an assessment of mechanical properties (YM & PR) to establish optimal formulations. Following this, cylindrical cement samples were prepared based on the optimal FA/ESP (75% + 25%) formulation, with concentrations of 0, 0.25, 0.50, 1, 2, and 4 BWOC (by weight of cement). The mechanical properties of these samples were then assessed. The Forney 920 Series machine is equipped with grips or fixtures appropriate for holding cylindrical samples to ensure that the sample is centered and aligned to avoid bending during the test. The machine uses the VFD controls, to apply a compressive load to the sample at a constant strain rate until it reaches the desired strain level or until failure occurs, it was automatically recorded the applied force and the corresponding deformation of the samples. The Forney 920 Series machine, equipped with appropriate grips or fixtures for holding cylindrical samples, ensured the samples were centered and aligned to prevent bending during the test. The VFD controls of the machine applied a compressive load to the sample at a constant strain rate until reaching the desired strain level or until failure occurred. The machine automatically recorded the applied force and the corresponding deformation of the samples throughout the test.

2.7. Compressive strength measurement

An investigation was conducted to understand the influence of fly ash and eggshell powder additives on cement slurry properties. A compressive strength assessment was done on various cement compositions. Compressive strength, which gauges the capacity of a cement

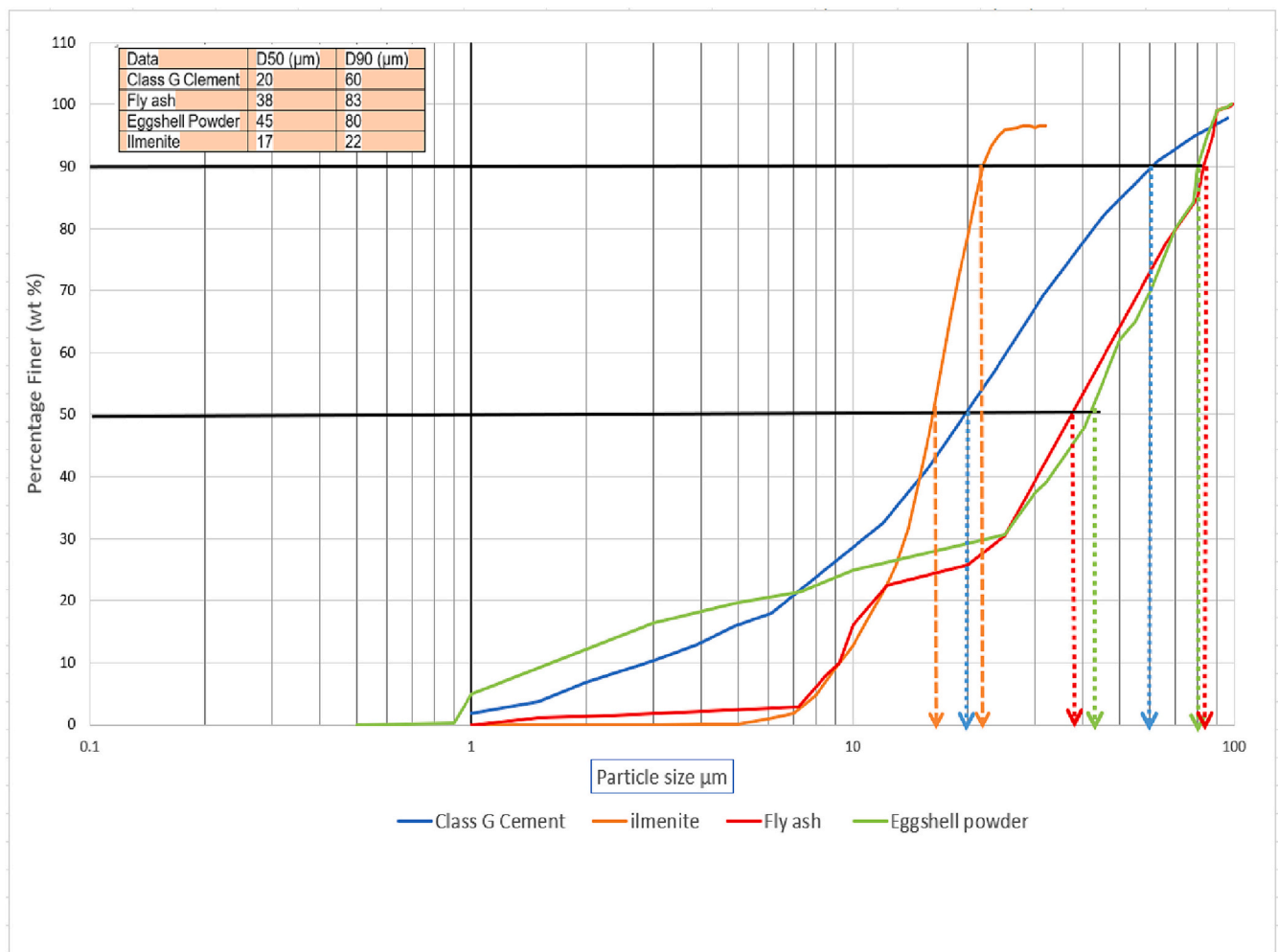


Fig. 6. The Particle size distribution of class G Cement, fly ash, eggshell powder, and Ilmenite.

mixture to withstand the maximum load before failure, was evaluated using cubic samples with dimensions of $2 \times 2 \times 2$ in.³. This method of testing is crucial for understanding the wellbore integrity of cement slurry when fly ash and eggshell powder is incorporated as an additive. II. High pressure and temperature curing accelerate cement hydration reactions and enhance compressive strength development compared to ambient curing [70]. After curing, a Forney 920 Series Universal Testing Machine with Automatic (VFD) Variable Frequency Drive controls was utilized to apply increasing compressive load on the cubes at a rate of 100 psi/s until failure occurred, per ASTM C39 procedures [71]. The maximum applied load at failure was used to calculate the compressive strength in psi. At least two replicate samples were tested for each cement formulation and the average strength recorded.

2.8. Tensile strength measurement

The study evaluates the effects of fly ash and eggshell powder on the properties of cement-based slurries. Cement formulations were prepared with varying amounts of Class F fly ash and eggshell powder incorporated as supplementary cementitious material. The tensile strength indicates the maximum tensile force a material can withstand prior to crack initiation and propagation. To quantify the tensile strength, samples were prepared from the cement mixtures according to [72]. This property is quantified by the maximum stress that the material can endure prior to failure when subjected to tensile forces. To ascertain this characteristic, cylindrical molds with a height of 4 in. and a diameter of 2 in. were used to cast the samples. The cement slurries were prepared

with water-cement ratios of 0.38, 0.44, and 0.5 and fly ash and eggshell powder contents of 0, 0.25, 0.5, 1.0, 2.0, and 4.0 wt% based on API RP 10B-2 standards [73]. Samples were cured at 100% humidity and 294 °F for 24 h per ASTM C1107 curing conditions [74]. These samples are then tested, using a Forney 920 Series Universal Testing Machine with Automatic (VFD) Variable Frequency Drive controls in accordance with the guidelines set forth by the ASTM International standards.

2.9. Porosity and permeability measurement

Porosity and permeability are critical parameters used to characterize the microstructure and fluid transport properties of cement matrices [73,17,75]. Porosity refers to the volume percentage of void spaces or pores within the cement sample. It is indicative of cement density and quality, with higher porosity generally linked to weaker mechanical strength [76]. On the other hand, permeability describes the interconnectivity of pores and ability for fluids to flow through the cement matrix. Permeability depends on pore sizes, pore throat diameters, and how well connected the pores are to each other [12]. A comprehensive cement evaluation involves determination of both porosity and permeability to assess density, microstructure, and fluid transport characteristics relevant for cement quality control and prediction of performance in downhole environments [77]. Cylindrical cement paste samples were prepared with dimensions of 4 in. (100 mm) length and 1.5 in. (38 mm) diameter. The samples specimens contained varying concentrations of fly ash, and eggshell powder by percent mass of cement (0, 0.25, 0.50, 1.0, 2.0, and 4.0% BWOC). The cement pastes

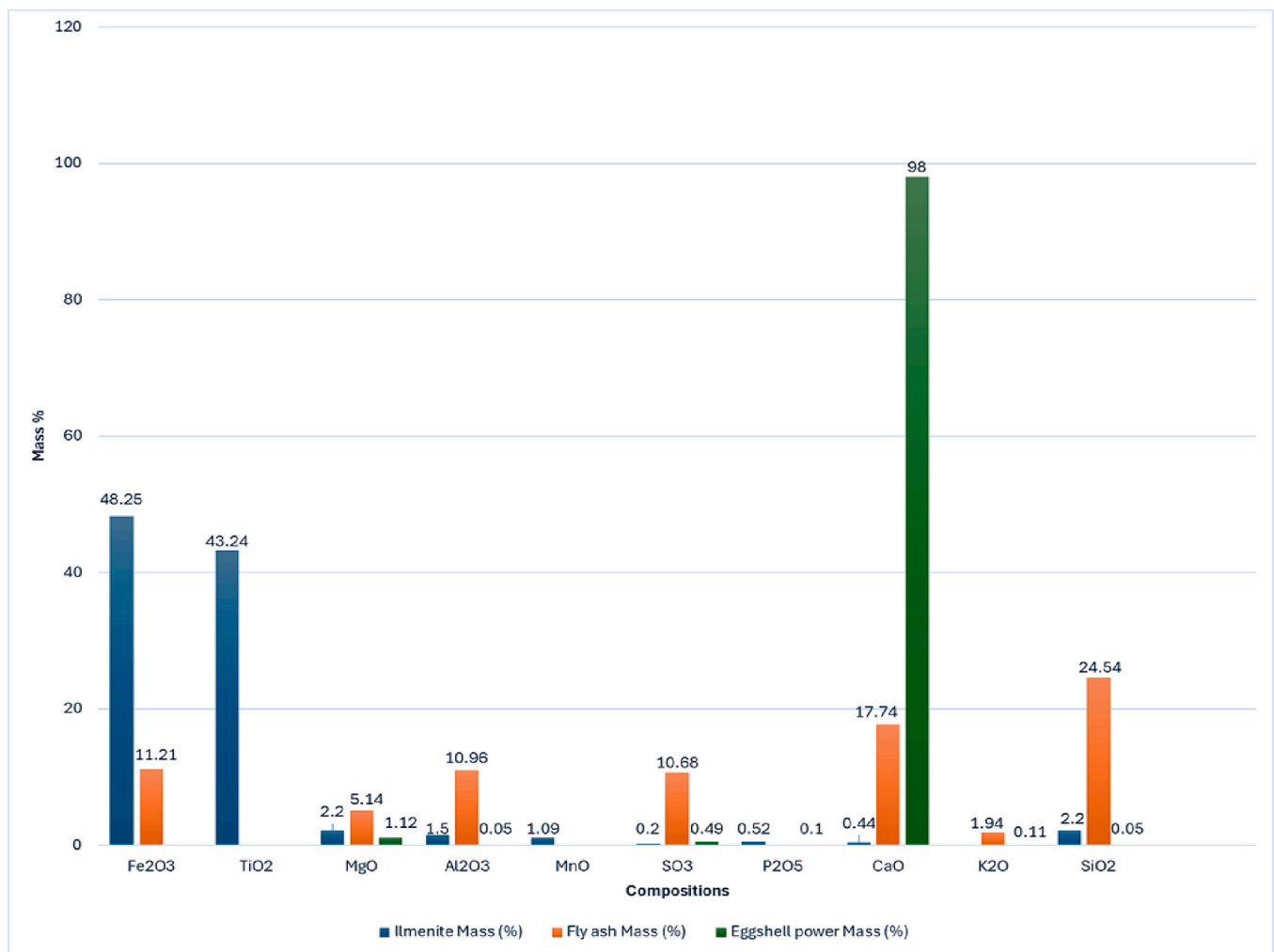


Fig. 7. X-ray fluorescence analysis of ilmenite, eggshell powder, and fly ash.

were mixed according to API RP 10B-2 [73]. The samples were subjected to a curing process under 100% humidity conditions at a temperature of 294 °F (145 °C) for 24 h based on ASTM C1107 standards (ASTM C1107–20, 2020). After curing, the cement samples were systematically evaluated to determine their porosity and permeability characteristics. The determination of the porosity of cement samples was conducted by employing a porosimeter based on the principles of Boyle's law [78]. The permeability of the cured cement samples was quantified using an Autolab-1500 multi-purpose test system in Fig. 5. The cement cores were mounted in a Hassler-type chamber and a confining pressure of 1000 psi was applied to prevent gas flow along the sample edges. Nitrogen gas was passed through the samples at regulated inlet pressures between 5 and 50 psi. For unsteady-state gas permeability measurement, the pressure pulse-decay technique coupled with the dual-chamber option for the AutoLab 1500 was utilized as described in ASTM D6391 (ASTM D6391–19, 2019). This involves applying an initial pressure pulse then monitoring the gradual decay. Permeability is derived by modeling the pressure-time relationship. The AutoLab 1500's sensitivity down to 0.15 psi differential pressure allows resolution of low permeability values down to the nanodarcy range for cements samples. All tests are performed under regulated confining pressures up to 1000 psi and temperatures up to 250 °F.

3. Results and discussion

3.1. Results on characterization of the materials and the formulated sample

Materials analyses were performed to determine the physical properties of Class G cement, fly ash, eggshell powder, and ilmenite, as detailed in Table 4. The specific gravity of these materials is as follows: Class G cement (3.12), fly ash (3.38), eggshell powder (2.5), and ilmenite (4.29). Particle size analysis conducted using Hydrometer analysis reveals that 50% of particles in Class G cement, fly ash, eggshell powder, and ilmenite are less than 20 µm, 38 µm, 45 µm, and 17 µm, respectively. Additionally, 90% of particles in these materials are smaller than 60 µm, 83 µm, 80 µm, and 22 µm, respectively. In Fig. 6, the particle size distribution analysis highlights the importance of fly ash and eggshell powder's median particle size (D50) and ilmenite's particle size below which 90% of particles fall (D90). These factors significantly affect the strength properties of cement when integrated into G class cement formulation as additives. Particle size distribution influences packing density, water demand, and the rate of pozzolanic reaction, thereby increasing the overall strength and durability of the cement [79,80]. The size of particles in fly ash additives significantly influences the strength properties of cement. Smaller particle sizes, indicated by lower D50 and D90 values in Table 4 and Fig. 6, increase the surface area available for the pozzolanic reaction. This reaction, occurring between the silica in fly ash and calcium hydroxide in cement paste, generates

additional calcium silicate hydrate (C-S-H), the primary binding agent responsible for concrete strength. Consequently, fly ash with smaller particle sizes can enhance both the strength and durability of cement by promoting a denser microstructure and filling voids within the cement paste [81,82]. XRF composition analyses are presented in Tables 5 and Fig. 7 for fly ash, eggshell powder, and ilmenite. Analysis of fly ash typically shows the presence of silicon, aluminum, iron, calcium, and other trace elements, which vary depending on the coal source and combustion conditions [83]. The main chemical component of fly ash is silica (SiO₂), followed by alumina (Al₂O₃) and iron oxide (Fe₂O₃) [84]. EDX analysis of eggshell powder reveals a high concentration of calcium, as calcium carbonate (CaCO₃) is the primary constituent of eggshells [85]. Other elements, such as magnesium, phosphorus, and potassium, may also be present in small quantities [86]. The microstructure and composition of fly ash and eggshell powder influence their performance as supplementary cementitious materials in concrete. The spherical shape and fine particle size of fly ash contribute to improved workability and reduced water demand in concrete mixtures [43]. The pozzolanic reaction between fly ash and calcium hydroxide (a byproduct of cement hydration) leads to the formation of additional calcium silicate hydrate (C-S-H) gel, which enhances the strength and durability of concrete [87]. Eggshell powder, being rich in calcium, can also participate in the hydration process and contribute to the formation of C-S-H gel [61]. Fig. 8 presents SEM imaging, detailing the morphology of eggshell, ilmenite, and fly ash particles, revealing their shape and surface structure. The irregular shape and rough surface of eggshell powder particles may improve the bonding between the cement paste and aggregates, leading to increased strength [88]. Furthermore, research on surface modification of fly ash through mechano-chemical treatment underscores that reducing particle size and altering surface morphology can notably improve fly ash's solubility in alkali solutions. This increased solubility indicates a more reactive pozzolanic material, suggesting that optimizing fly ash particle size through treatments can enhance its reactivity and its contribution to cement strength properties [89]. Eggshell powder, primarily composed of calcium carbonate, also impacts cement strength properties through its particle size. Like fly ash, finer particle sizes of eggshell powder offer a larger surface area for chemical reactions with cement paste. Calcination of eggshell powder can result in the formation of reactive lime, which further engages in hydration reactions, aiding in cement strength development. Additionally, the mechanical properties of eggshell-derived materials, such as toughness and crack-bridging capabilities, though not linked to cement, suggest that finely ground eggshell powder could potentially enhance the microstructural integrity of cementitious composites [90].

3.2. Microstructure analysis of fly ash and eggshell powders

One very effective analytical technique for analyzing the composition and microstructure of cementitious materials is scanning electron microscopy (SEM). SEM makes it easier to identify various phases and constituents that are embedded in the cement matrix through providing

detailed surface morphology images [15]. Analyzing the microstructure of FAESP (with optimal fly Ash and eggshell powder of 2% BWOC) samples in Fig. 9 reveals a heterogeneous matrix made mostly of cement hydration products with an uneven shape and no spherical particles. The FAESP sample's SEM analysis reveals a tightly packed microstructure linked with a strong bonding network. The images clearly show the densely packed and networked structure of the FAESP sample, indicating excellent binding between cement component, fly ash and eggshell powder particles, with nano-fractures range of 726.4 nm (0.726 μ m) to 863.8 nm (0.864 μ m). This small microstructure, strengthened by a strong bonding network, has the potential to improve mechanical characteristics and increase the performance of the sample with optimal ratio of FAESP compared to the base sample which has notable micro-scale fractures of 3.076 μ m (3076 nm) and 3.731 μ m (3731 nm) inside its framework in a various application as demonstrated in Fig. 10. SEM imaging analysis of the base sample reveals a less dense, less compact nature and more porous structure and microfractures, though in a less condensed setting, may affect the geo-mechanical properties and performance of the base sample.

3.3. Analysis of the chemical composition of the materials and the formulated samples

Energy Dispersive X-ray Spectrometry (EDX) analysis provides information about the elemental composition of various materials. Figs. 11 and Fig. 12 show the chemical composition of FAESP and BS respectively which was examined using Energy Dispersed X-ray Analysis (EDX). The analysis revealed peaks of elements present in the FAESP sample, with calcium (Ca) being the most abundant, exhibiting a concentration mass of 40.07%. This high calcium content can be attributed to the presence of eggshell powder, which is primarily composed of calcium carbonate (CaCO₃) [55]. In contrast, the base sample (Class G cement) shows a lower calcium content of 29.71%. The significant amount of calcium in the FAESP sample suggests its potential as a supplementary cementitious material, as calcium plays a crucial role in the hydration and strength development of cementitious systems [12]. Oxygen (O) is present in both samples, with a concentration mass of 22.72% in the FAESP sample and 34.43% in the base sample. The higher oxygen content in the base sample can be attributed to the presence of oxidized compounds in the cement, such as calcium oxide (CaO) and silicon dioxide (SiO₂) [45,13]. Silicon (Si) and aluminum (Al) are also present in the FAESP sample, with concentration masses of 16.67% and 1.78%, respectively. The silicon content in the FAESP sample, although lower than the base sample, still indicates the presence of pozzolanic materials from fly ash, which can contribute to the formation of additional calcium silicate hydrate (C-S-H) gel, thus enhancing the strength and durability of cementitious systems [87,43]. Interestingly, the FAESP sample contains a significant amount of titanium (Ti), with a concentration mass of 9.18%, while the base sample has a titanium content of 29.71%. The presence of titanium in fly ash can be attributed to the coal source and the combustion conditions in power plants [83]. Titanium dioxide (TiO₂) has been reported to have photocatalytic properties and can contribute to the self-cleaning and air-purifying capabilities of cementitious materials [92]. Iron (Fe) is present in both samples, with a concentration mass of 8.78% in the FAESP sample and 3.17% in the base sample. Iron is commonly present in fly ash, typically in the form of iron oxides (Fe₂O₃ and Fe₃O₄) [93]. The presence of iron can influence the color and magnetic properties of the FAESP sample [94]. Additionally, iron oxides have been reported to contribute to the hydration process and enhance the compressive strength of cementitious materials [95]. Aluminum (Al) is detected in both samples, with a concentration mass of 1.78% in the FAESP sample and 1.37% in the base sample. Aluminum, along with silicon, is a key component of the aluminosilicate phases present in fly ash, contributing to its pozzolanic reactivity [45]. Sulfur (S) is present in the FAESP sample with a concentration mass of 0.84%, while it is not detected in the base sample. The presence of sulfur in the

Table 5

Show chemical compositions of ilmenite, fly ash and eggshell powder.

Component	Ilmenite Mass (%)	Fly ash Mass (%)	Eggshell powder Mass (%)
Fe ₂ O ₃	48.25	11.21	–
TiO ₂	43.24	–	–
MgO	2.20	5.14	1.12
SiO ₂	2.19	24.54	0.05
Al ₂ O ₃	1.48	10.96	0.05
MnO	1.09	–	–
SO ₃	0.2026	10.68	0.49
P ₂ O ₅	0.52	–	0.1
CaO	0.44	17.74	98
Na ₂ O	–	15.84	–
K ₂ O	–	1.94	0.11

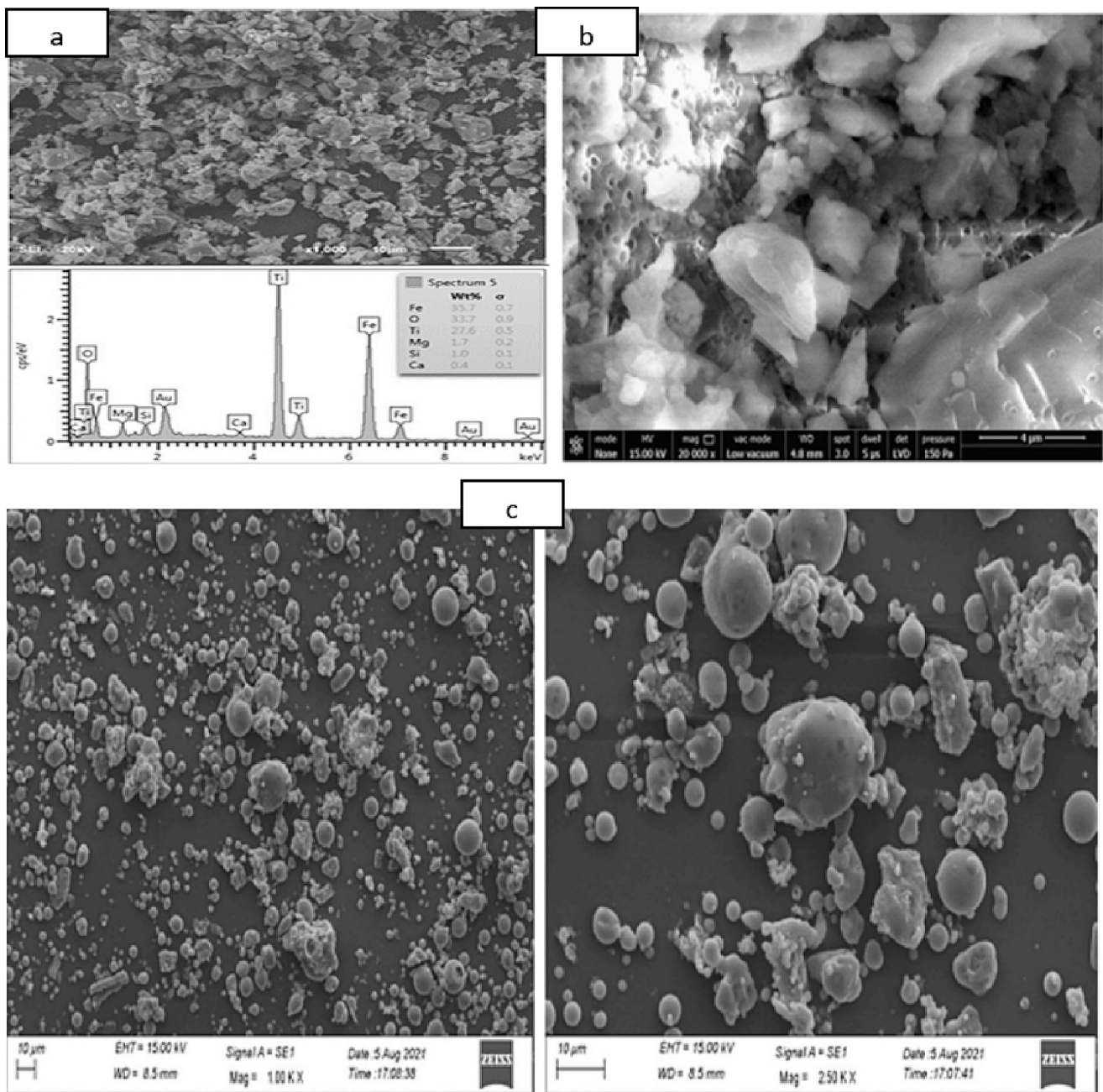


Fig. 8. SEM Morphology of raw material of (a) Ilmenite, (b) Eggshell powder and (c) Fly ash ([64,25,91]).

FAESP sample may be attributed to the sulfur-containing compounds in coal, such as pyrite and organic sulfur [96]. Notably, lead (Pb) is detected in the base sample with a concentration mass of 2.82%, while it is not present in the FAESP sample. The presence of lead in the base sample may raise concerns regarding the environmental impact and potential leaching of heavy metals from the cement [97]. The absence of lead in the FAESP sample suggests that it may be a more environmentally friendly alternative to traditional cement. The EDX analysis provides valuable insights into the elemental composition of the FAESP sample, highlighting the predominance of calcium, oxygen, silicon, titanium, iron, aluminum, and sulfur. The high calcium content from eggshell powder and the presence of pozzolanic elements from fly ash underscore the potential of FAESP as a supplementary cementitious material. The titanium content may impart additional photocatalytic properties, while the iron content can influence the color and magnetic properties of the material. The sulfur content, although relatively low,

may have implications for the setting and durability of cementitious systems.

3.4. Effect of optimal concentration of fly ash and eggshell powder on the rheological characteristics of cement slurry

The rheological characteristics of a cement slurry used in oil and gas wells and underground gases storage wells are critical parameters that influence the slurry's composition and performance, ensuring enhanced functionality and determining the feasibility of pumping the slurry into the wellbore. Fig. 13 shows the determination of the optimal FAESP concentration ratio. The figure illustrates the changes in rheological behavior of the cement slurry as the proportion of fly ash and eggshell powder is altered. As depicted in Fig. 13a, an initial decline in plastic viscosity (Base case) is noted with the incremental addition of fly ash and eggshell powder, followed by a rise started from proportional ratios

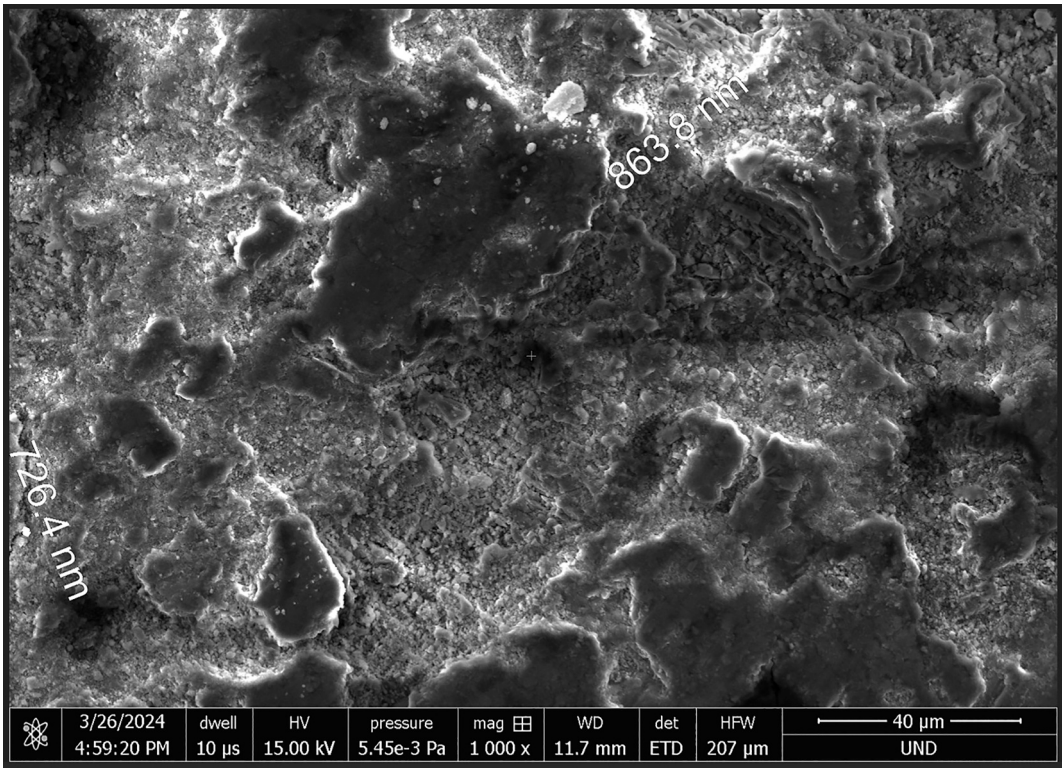


Fig. 9. SEM Analysis of FAESP samples.

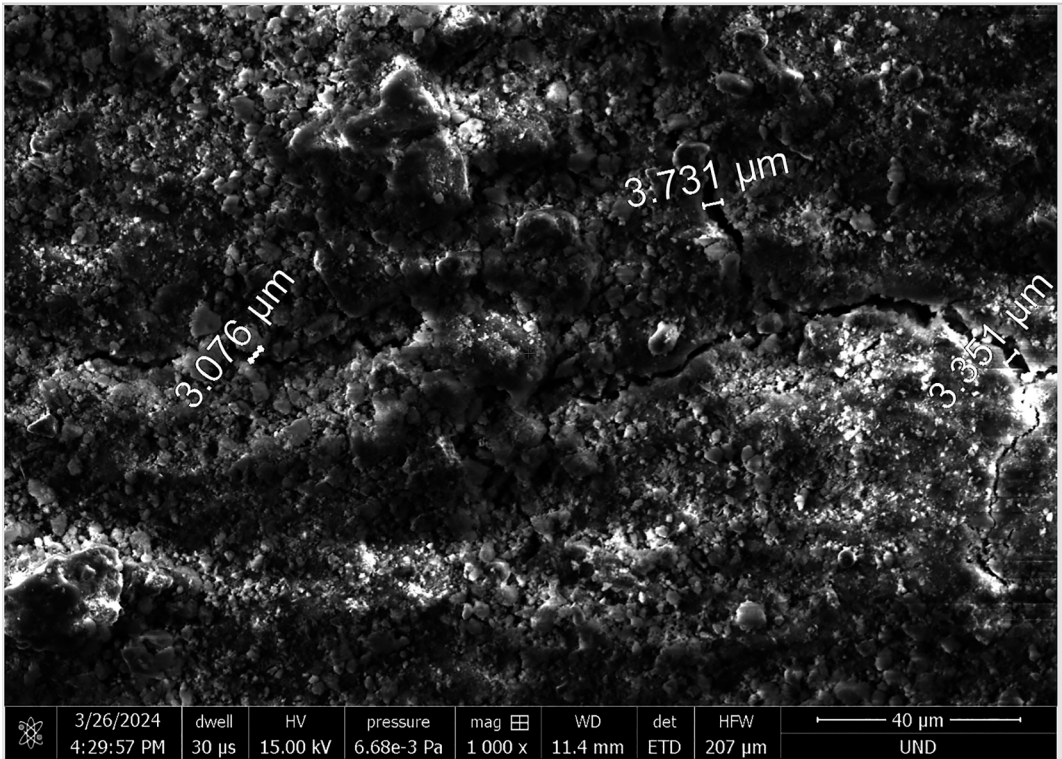


Fig. 10. SEM Analysis of Base samples (BS).

of FAEPS by weight of cement (BWOC). Fig. 13a established FAESP (75%/25%) ratio concentration as an optimal FAESP ratio concentration with plastic viscosity of 255.5 (Ib/100) ft². This specific ratio was identified as the most conducive for achieving the desired plastic

viscosity in the final cement slurry formulation due to its ability to ensure optimal flow characteristics for high-performance cement applications [98]. As depicted in Fig. 13b, a yield point showed an overall increasing trend with rising FAESP ratio concentration and suddenly

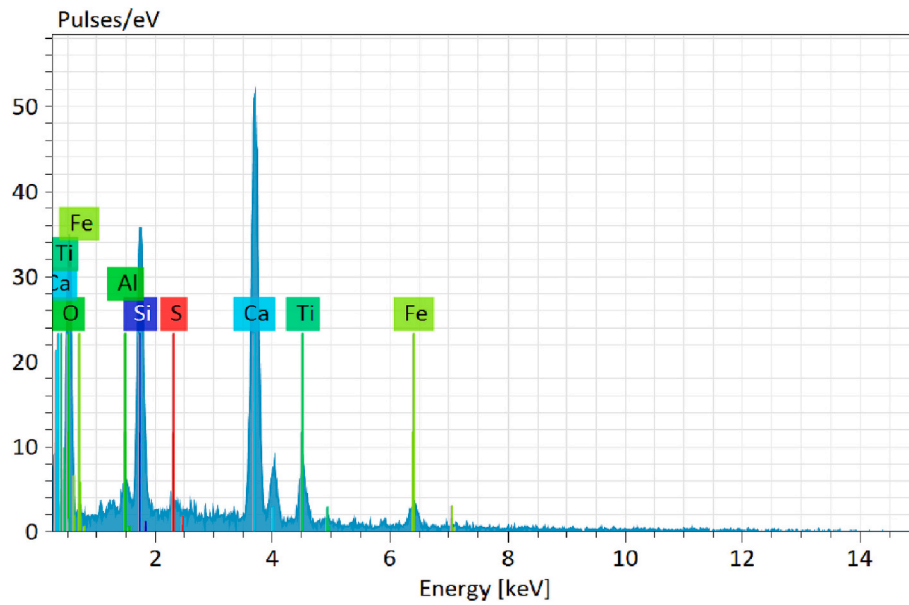


Fig. 11. EDX Analysis of FAESP sample.

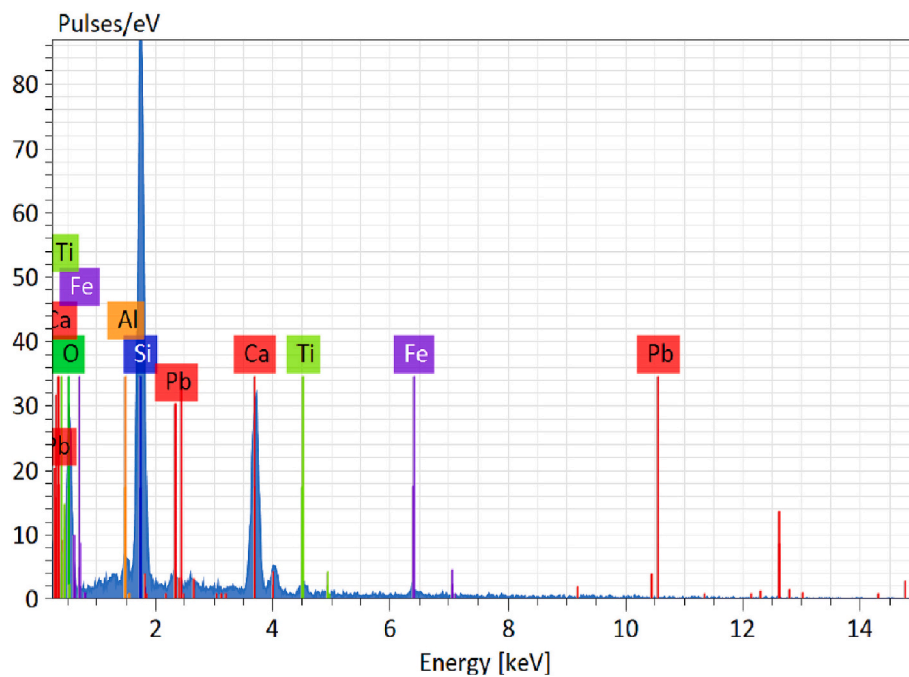


Fig. 12. EDX Analysis of BS sample.

decrease in FA/ESP (75% + 0%) by weight of cement (BWOC). As described by Clark et al. [99], yield stress and plastic viscosity are inversely related to a given shear stress and rate for real systems. Aligning with this relationship, the yield point and plastic viscosity showed opposite tendencies in the present study. Fig. 13b establishes FA/ESP concentration ratio at 75%/25% as the optimal proportion with a yield point of 40 (lb/100) ft.² for the final cement slurry formulation, specifically tailored to enhance the yield point with its ability to ensure the slurry's robustness in withstanding applied stresses. Fig. 13c show the 10-s gel strength followed a similar pattern to the yield point, rising with incremental in FAESP ratio concentrations. Fig. 13c established that the combination ratio of 75%/25% (FAESP) conferred the maximum 10-s gel strength of 10.25 (lb/100) ft.² among the FAESP mixtures evaluated, indicative of optimal rheological enhancement. This

optimal FAESP concentration was selected for the final cement slurry formulations subjected to mechanical property assessments.

3.5. Effect of fly ash and eggshell powder on the compressive strength of cement slurry

The compressive strength of cement formulations is a key consideration when designing wellbore cementing operations. This study examined the compressive strengths of slurry mixtures with varying fly ash and eggshell powder concentrations to established which ratio concentrations' given the best compressive strengths for the final cement slurry formations. Fig. 14 shows an increase in compressive strength with an increase in proportional ratios of FAESP by weight of cement (BWOC) and it has shown that a FAESP ratio of 75%/25% has been

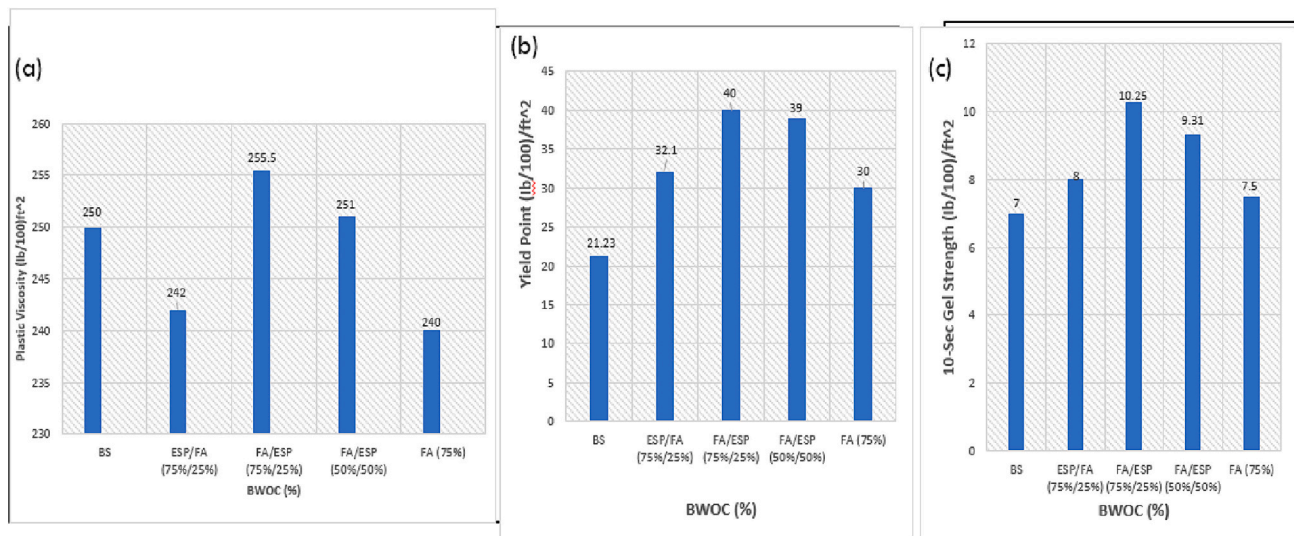


Fig. 13. Effect of optimal fly ash and eggshell powder concentration on rheological properties of oil well cement slurry; (a) Plastic Viscosity; (b) Yield Point; (c) 10-s Gel Strength.

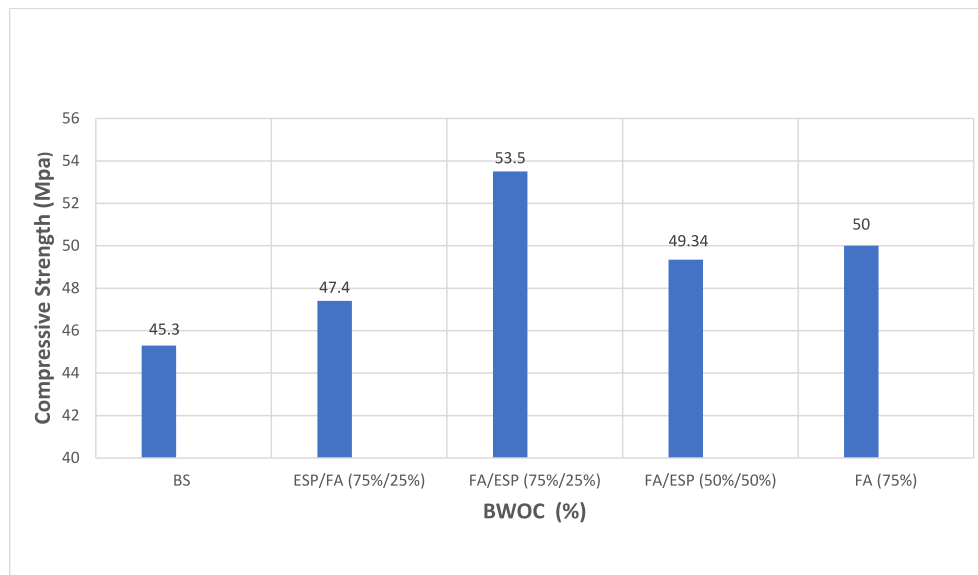


Fig. 14. Effect of optimal fly ash and eggshell powder particles on the strength of oil well cement for Compressive strength.

identified as the optimal concentration, yielding a compressive strength of 53.5 MPa. This optimal synergistic blend, which valorizes industrial byproducts, demonstrates a remarkable 18.10% enhancement in performance compared to standard plain cement (BS), affirming its viability as a sustainable and superior cementitious matrix.

3.6. Effect of fly ash and eggshell powder on the tensile strength of cement slurry

The tensile strength of cement formulations is a critical factor in the engineering of wellbore cementing processes. This study initially investigates the tensile strength of slurry blends with differential proportions of FAESP with the aim of establishing which ratio concentrations' given the best tensile strengths for the final cement slurry formations. Fig. 15 show an increase in tensile strength with an increase in proportional ratios of by weight of cement (BWOC), the optimal 75% /25% (FA/ESP) concentration ratio by weight exhibited the maximum tensile strength among the different formulations

examined, achieving 13.53 MPa. This optimal synergistic blend of industrial byproduct valorization showcases 318.9% enhancement versus plain cement (BS), substantiating its potential as a sustainable advanced cement matrix.

3.7. Effect of fly ash and eggshell powder on the elastic properties of cement slurry

The mechanical resilience of cementitious materials is pivotal for ensuring the enduring structural soundness of wellbores for oil and gas operations or underground gases storage (H₂ & CO₂). Cement formulations characterized by a reduced modulus of elasticity and an elevated Poisson's ratio are typically associated with enhanced pliability and an augmented capacity to withstand shear-induced deformations. Fig. 16 illustrates the impact of FAESP addition on the elastic performance with the aim of establishing the optimal ratio concentration for the final cement slurry formation. As shown in Fig. 16a, Young's modulus increased initially to the maximum of 75%/25% FAESP with in an

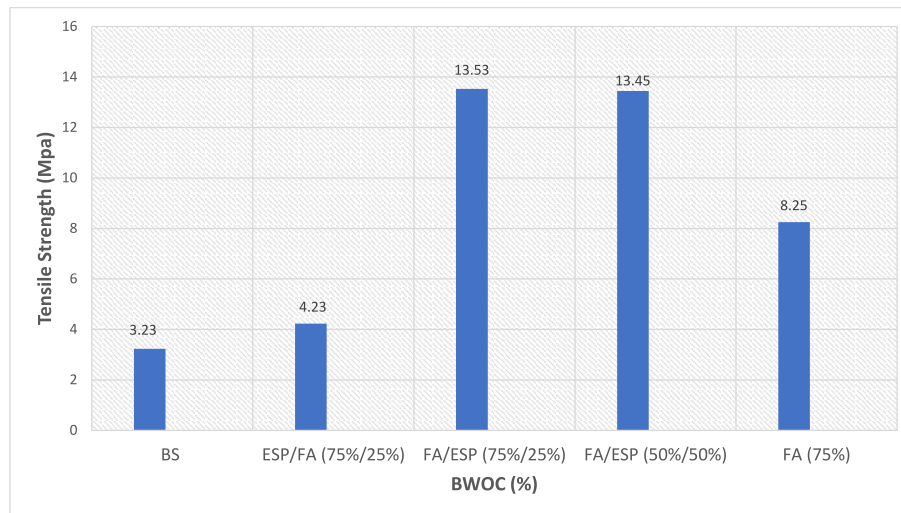


Fig. 15. Effect of optimal fly ash and eggshell powder particles on the strength of oil well cement for Tensile strength.

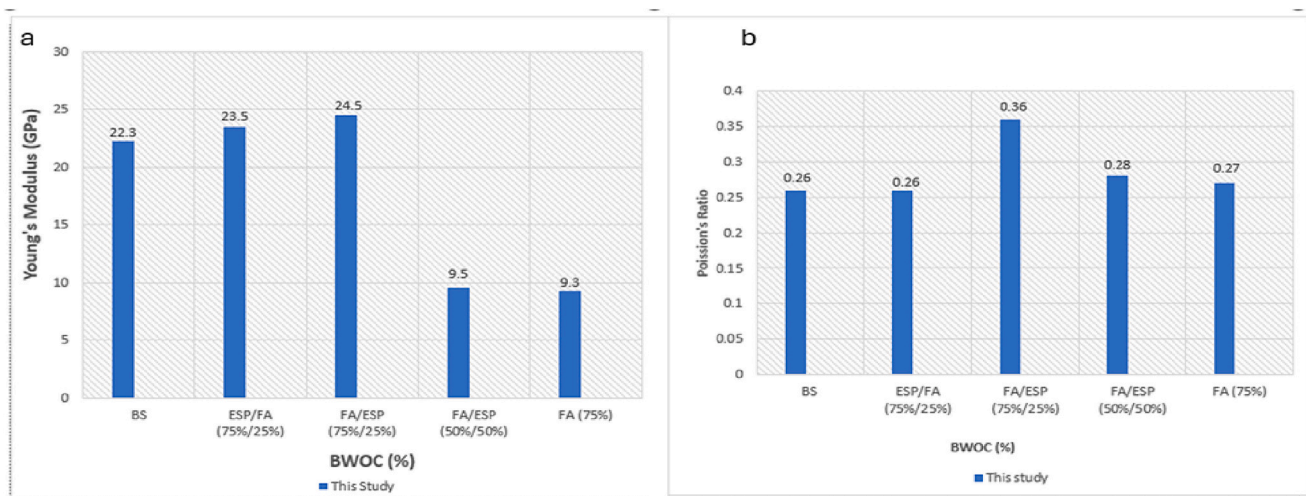


Fig. 16. Effect of optimal fly ash and eggshell powder particles on the elastic properties of oil well cement; (a) Young's Modulus; (b) Poisson's Ratio.

increase in proportional ratios FAESP, achieving 24.5 MPa increase with base cement (BS), before it decreases again with the concentration ratios by weight of cement (BWOC), confirming 75%/25% FAESP ratio concentration as the optimal for the final cement slurry formulation. Fig. 16b. presents the effect on Poisson's ratio with the concentration ratios of FAESP. Fig. 16b, initially shown similar increases in Poisson's ratio with a proportional ratio of FAESP, before increase to the maximum of 75%/25% FAESP with in an increase in proportional ratios FAESP, achieving 0.36 poisson's ratio increase with base cement (BS). This optimal synergistic incorporation, which valorizes industrial byproducts, show a significant 38.46% enhancement in performance compared to standard plain cement (BS), affirming it as the optimal ratio for the final cement slurry formulation.

3.8. Effect of optimal FAESP ratio concentration on the rheological characteristics of the final cement slurry formulation

The rheological characteristics of a cement slurry used in oil wells and underground gases storage (H₂ & CO₂) are critical parameters that influence the slurry's composition and performance, ensuring enhanced functionality and determining the feasibility of pumping the slurry into the wellbore. The combined use of FAESP in cement slurry presented a complex interaction between the two additives. The addition of fly ash

to the cement slurry was found to influence the rheological properties significantly. Consistent with previous findings, the incorporation of fly ash reduced the yield stress and plastic viscosity, indicating an improvement in the workability of the cement slurry [39], for eggshell powder, the filling effect and the presence of fine particles contributed to an increase in the viscosity of the slurry. The balance between these opposing effects was critical in achieving the desired rheological characteristic by ensuring good pumpability without compromising stability. Key rheological factors such as plastic viscosity, yield strength, and gel strength at 10 s were determined by recording the shear stress at varying shear rates using a viscometer at a temperature of 194 °F. A lower plastic viscosity is desirable as it directly impacts the pressure required for pumping. Conversely, a higher yield point is advantageous as it enhances the slurry's capacity to transport materials. Furthermore, a greater gel strength is beneficial as it plays a role in managing gas migration within the slurry [100,101,102]. Fig. 17a as the final formulation shows the influence of increasing FAESP (75%/25%) concentration on the rheological properties of the cement slurries. It can be observed that the plastic viscosity initially decreased with FAESP addition up to 0.5% BWOC, but a slight increase as compared to the previous study [103]. As the FAESP content was further increased, the plastic viscosity rose, reaching a maximum at 2% BWOC addition. This viscosity increase may arise from the higher water absorption capacity of

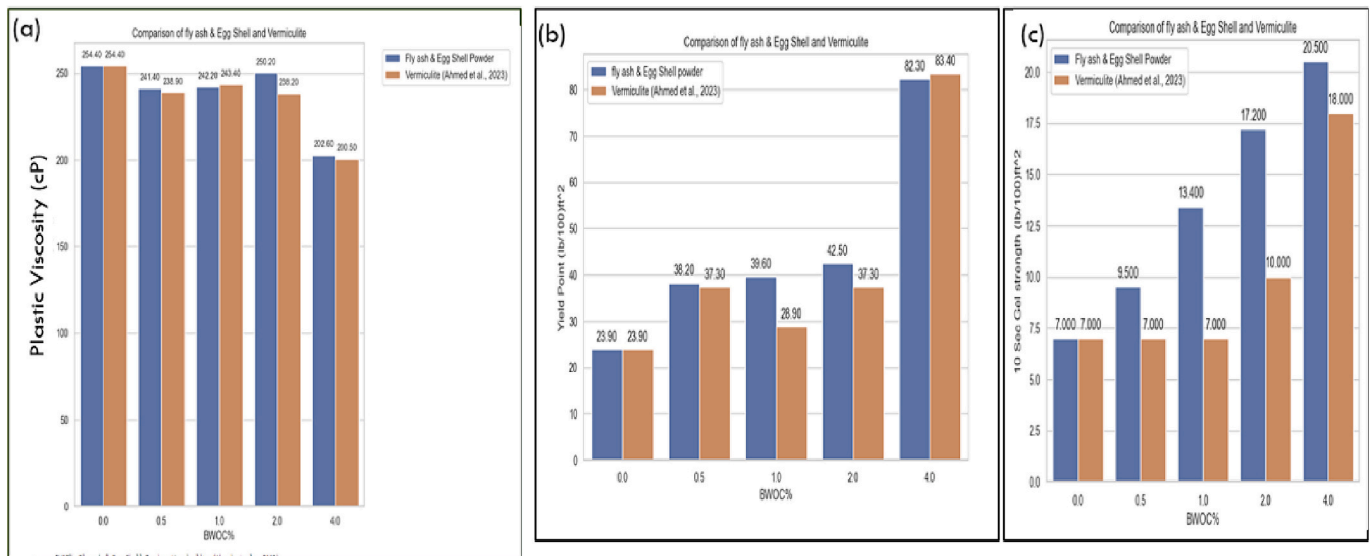


Fig. 17. Effect of final formulation of fly ash and eggshell powder particles on rheological properties of oil well cement slurry; (a) Plastic Viscosity; (b) Yield Point; (c) 10-s Gel Strength.

fly ash and eggshell powder, reducing the free water content and increasing the solid fraction, which can elevate the plastic viscosity [104], but sharply increase as compared to the vermiculite studied [105]. Compared to the prior vermiculite-modified system [103], the 75/25 fly ash/eggshell powder (FAESP) cement blend improves plastic viscosity achieving 250.20 cP. For cementing casing strings in underground gas storage wells, adequate viscosity and pumpability facilitate injectability into narrow annular spaces, efficient slurry placement for optimum zonal isolation. This final optimal formation concentrations at 2% BWOC allows trouble-free wellbore flow, even within narrow annuli or challenging inclined/horizontal trajectories. This ensures consistent, hole-conforming slurry emplacement to minimize channeled flow or contamination risks that undermine long-term barrier functionality. Moreover, the heightened viscosity provides superior suspension capacity to prevent slurry stiffening/gelation, particle sedimentation and dehydration issues during static pumping times. Also, higher plastic viscosity helps stability during curing and reduced fluid loss. As described by Clark et al. (1990), yield stress and plastic viscosity are inversely related to a given shear stress and rate for real systems. Aligning with this relationship, the yield point and plastic viscosity showed opposite tendencies in the present study. Fig. 17b as the final formulation shows an increment in FAESP (75% + 25%) concentration on the rheological properties of the cement slurries. It can be observed that the yield point increases with FAESP addition up to 2% BWOC, but a slight decrease as compared to the vermiculite study [103]. As the FAESP content of 2% BWOC was further increased, the yield point rose, reaching a maximum at and 4% BWOC addition. Relative to the prior vermiculite system [103], the optimized 75%/25% FAESP cement blend enhances yield stress achieving 42.50 lb./100 ft.². In underground gases storage wells, an adequate yield point prevents pressure-driven slurry infiltration into the formation while enabling the cement slurry displacement. Also, improved yield strength helps the cement matrix to withstand larger deformations prior to physical breakdown, improve suspension of particulates in the slurry, reduce segregation and sedimentation, enhance stability of the slurry, boosting long-term durability to maintain reliable zonal isolation over wells lifetimes spanning decades. However, the highest yield point recorded at 4% BWOC is not considered because the cement slurry has more viscous that requires greater force to initiate flow. This led to difficulties in pumpability of the cement slurry down the wellbore and into the annular space between the casing and the formation. It also decreases the flowability of the cement slurry into the narrow spaces and micro-annuli that can exist between

the casing and the formation. Fig. 17c as the final formulation shows an increment in FAESP (75%/25%) concentration on the rheological properties of the cement slurries. It can be observed that the 10-s gel strength increases with FAESP concentration at the maximum of 2% and 4% BWOC, but with a significant increase. The optimal 75%/25% fly ash/eggshell powder (FAESP) cement demonstrated over 17.7% higher 10-s gel strength than the previous vermiculite composite [103], boosting static stability. The enhanced 10-s gel strength observed in this study confers several advantages for applications in underground gas storage and oil and gas well operations. These benefits include improved wellbore stability, augmented suspension of cuttings and weighting materials, and superior sealing capabilities to mitigate fluid loss. Additionally, the increased gel strength contributes to the maintenance of a consistent hydrostatic pressure within the wellbore, thereby reducing the risk of formation fluid influx and ensuring safer well completions practices. High gel strengths at 4% BWOC is not selected because its lead to increase in viscosity and frictional pressure losses, making slurry pumping and placement more difficult, while also narrowing down placement time windows before unmanageable levels of gelation. There is thus an optimum ceiling beyond which further rise in static gel strength becomes counterproductive by hindering rather than bolstering cementing reliability. The addition concentration of 2% of FAESP in the cement slurry considered to be the optimum concentration for better rheological properties with the low plastic viscosity, high yield point and high gel strength. However, concentration of 4% BWOC FAESP is not considered to be optimal, as it leads to excessive elevation of the yield point and gel strength that could detrimentally impact cement slurry pumpability.

3.9. Effect of optimal FAESP ratio concentration on the compressive strength of the final cement slurry formulation

The compressive strength of cement formulations is a key consideration when designing wellbore cementing operations. This study examined the compressive strengths of slurry mixtures with varying fly ash and eggshell powder concentrations. The pozzolanic activity of fly ash and the nucleation sites provided by eggshell powder could potentially enhance the microstructure and mechanical properties of the cement matrix [106,107]. As shown in Fig. 18, the compressive strength increases with increasing of the optimal FAESP (75%/25%) content, reaching a maximum value at 2% FAESP by weight of cement (BWOC) as the best compressive strength, before decreasing again at higher

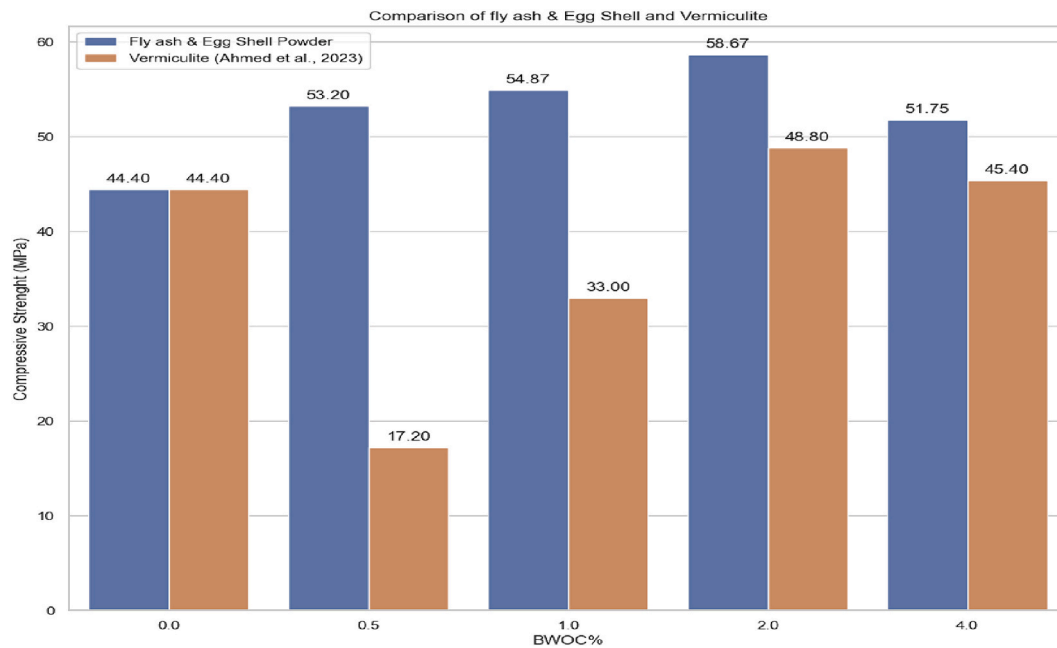


Fig. 18. Effect of optimal ratio of fly ash and eggshell particles on the strength of oil well cement for Compressive strength.

concentrations. Notably, the compressive strength at 2% BWOC FAESP exceeded that of the plain cement slurry without FAESP. In this present investigation, the optimized formulation of FAESP, comprising a 75%/25% concentration ratio, exhibited superior efficacy performance as compared to the vermiculite-based composition previously reported in Ahmed et al. [103]. Compared to the prior vermiculite-modified cement system [105], the present study's optimal combination of FAESP 75%/25% by weight achieves a 32.13% improvement in compressive strength, reaching 58.67 MPa versus 44.4 MPa. The subsequent increase in compressive strength with 2% of FAESP concentration can be explained by enhanced hydration and changes in the cement matrix microstructure. Specifically, the FAESP appears to fill voids and undergo thermal expansion, producing a denser cement system with improved compressive strength [107]. The present study in Fig. 15 showing higher compressive strength than previously studied vermiculite [103] helps to ensure the longevity and durability of the storage or well structure, reducing the likelihood of structural failure over time or mechanical stress that occur due to the high pressures, high temperature and corrosive fluids usually found in underground gas storage and oil and gas wells [108]. Moreover, it also helps to significantly enhance the load-bearing capacity which directly translates to enhanced wellbore integrity and safer, more reliable long-term underground gas storage in depleted reservoirs. The high-strength matrix provides critical anchoring for casing strings and an augmented barrier to leakage under stresses exerted by pressurized hydrocarbons. The present study recording 2% of superior cement toughness resists propagation of micro-cracks that compromise infrastructure resilience as compared to the previously studied vermiculite [103].

3.10. Effect of optimal FAESP ratio concentration on the tensile strength of the final cement slurry formulation

The tensile strength of cement formulations is a key consideration when designing wellbore cementing operations. This study examined the tensile strengths of slurry mixtures with varying of optimal 75%/25% FAESP concentrations from 0.25% to 4% by BWOC. The impact of FAESP on the tensile strength of cement slurry can be inferred from the effects these materials have on the hydration process and the microstructure of the cement matrix. Fly ash, when added to cement slurry, has been shown to alter the hydration kinetics, and microstructure of the

resulting composite material [109]. Eggshell powder, on the other hand, has been found to accelerate the hydration reaction of cement paste and influence the formation of hydration products [110]. Beyond superior compressive strength, the final formulation of 75%/25% FAESP at 2% of cement matrix also achieves a remarkable 12.05 MPa improvement in tensile strength compared to the prior vermiculite system [105]. Also, the tensile strength gains significant improvement of 215.45% at 2% BWOC compared to the base sample. This has profound help for long-term durability under cyclic loading of underground gases storage and hydrocarbon production wells due to the injection and withdrawal of the gases. The high tensile capacity in Fig. 19 provides a crucial safeguard against debonding, induces thermal expansion and contraction, micro annulus formation, and mechanical damage-driven leaks. The high interface bond strength and matrix cohesion hence preserve zonal isolation integrity to prevent risks of well failures linked to induced seismic events, or operational stresses. This study's sustainable blended cement slurry design enhances resilience to a broad suite of downhole threats that facilities can encounter over decades of operations. Moreover, the FAESP formulation's superior tensile strength equips it to withstand the aggressive temperature fluctuation and chemical conditions prevalent in deep geological environments. Such robustness is vital for maintaining a strong seal of storage wellbores, safeguarding against gases and fluid leakage and the consequent environmental issues.

3.11. Effect of optimal FAESP ratio concentration on the elastic properties of the final cement slurry formulation

The elastic properties of cement are critical for long-term wellbore integrity. Cements with lower Young's modulus and higher Poisson's ratio generally exhibit better ductility and resistance to shear failures ([111]; [112]). As shown in Fig. 20a, the Young's modulus increases with increasing in FAESP 75%/25% concentration, reaching a maximum value at 1% FAESP by weight of cement (BWOC) before decreasing again sharply at 2% BWOC as compared to the vermiculite studied [103] where there is increase at 2% BWOC. However, Young's modulus with lower performance at 2% by BWOC of FAESP concentration performance better than higher performance at the 1% BY BWOC of FAESP concentration. Young's modulus depends on factors like moisture content, aggregate volume, density, and strength [113]. Fly ash can improve the workability of cement slurry and reduce the water demand, which

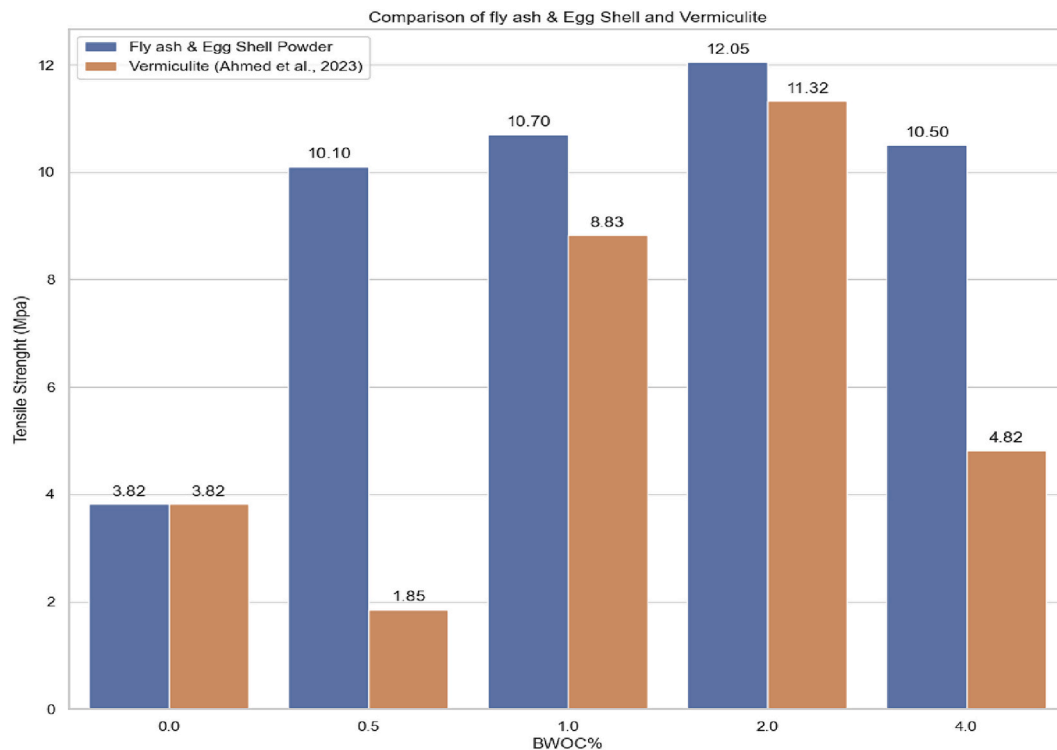


Fig. 19. Effect of final formulations fly ash and eggshell powder particles on the strength of oil well cement for Tensile strength.

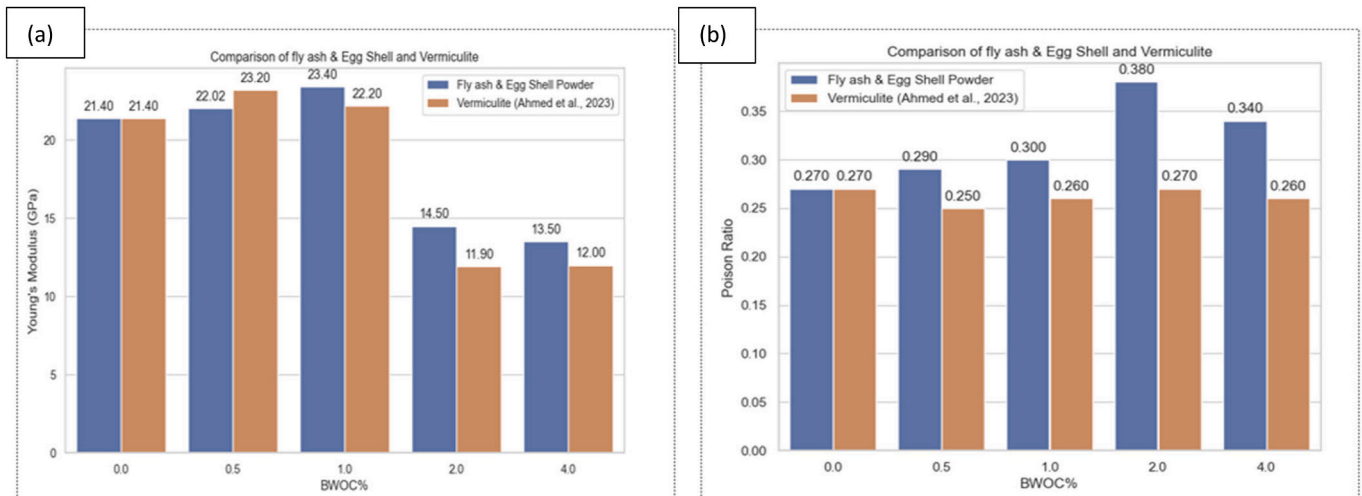


Fig. 20. Effect of final formulations of fly ash and eggshell powder particles on the elastic properties of oil well cement; (a) Young's Modulus; (b) Poisson's Ratio.

may lead to a denser microstructure upon hydration and potentially increase the modulus of elasticity [39]. Notably, 2% BWOC of FAESP with reduced Young's modulus exhibits greater ductility and fracture toughness, reduced vulnerability to shrinkage cracking, enhanced seismic performance, resisting formation of microcracks that compromise zonal isolation of underground gases storage or oil and gas wells as compared to the vermiculite studied [103] with high Young's modulus. This preserves long-term barrier functionality amidst shifting loads, subsurface shifts, or well interventions. As shown in Fig. 20b, the Poisson's ratio increases with increasing in FAESP 75%/25% concentration, reaching a maximum value at 2% FAESP by weight of cement (BWOC) as compared to the previous study [103]. Complementing the superior cement elasticity demonstrated at 2% BWOC via reduced Young's Modulus, the 75%/25% FAESP concentration achieved at 2% BWOC

higher Poisson's ratio as compared to the vermiculite-modified matrix by Ahmed et al. [103]. In underground gases storage and gas/oil wells applications, this enhanced lateral strain capacity and resilience translates directly into durable long-term wells integrity. Moreover, this also helps to boost incompressibility, enhance crack resistance, reduce stress concentration, potentially enabling better casing standoff and withstand heavy stresses over the well's life cycle compared to the previous studied vermiculite [103] with lower Poisson' ratio of 0.27.

3.12. Effect of optimal FAESP ratio concentration on porosity and permeability of the final cement slurry formulation

The porosity and permeability of cement are critical petrophysical parameters that govern wellbore durability and fluid migration behind

casings over the production lifetime [114]. Proper evaluation of these properties in cement formulations is essential to prevent additional remedial cementing operations, which add costs and can risk well integrity. This study examined the impact of fly ash and eggshell powder concentration on cement porosity and permeability. The decrease in porosity at higher FAESP loadings can be attributed to enhanced hydration and microstructural refinements [115]. Fig. 18a the cement sample shows a better porosity at 2% BWOC with the final optimal concentration ratio of 75%/25% (FAESP). Fig. 18a shows that cement sample permeability decreases slightly upon initial in FAESP (75%/25%) incorporation up to 1% by weight of cement (BWOC). With further in FAESP (75%/25%) additions, permeability was reduced, reaching an optimum value at 2% FAESP by weight of cement (BWOC). The 75%/25% FAESP cement matrix reduces permeability by over 40% compared to the vermiculite formulation [103], reaching 0.012 mD. Restricting interconnected flow paths helps to improve the containment of gases and reduces the risk of leakage and environmental contamination from underground gas storage wells. The reduced permeability hence preserves long-term well integrity and containment efficiency even after decades of service. As depicted in Fig. 21, the optimized cement formulation containing 75% fly ash and 25% eggshell powder (FAESP) by BWOC demonstrates improved porosity at 2% FAESP loading compared to the plain oil well cement (BS). Compared to the prior vermiculite-based system [103], the optimized final formulation 75%/25% FAESP cement matrix reduces porosity, a remarkable enhancement in durable zonal isolation capacity pivotal for underground gas storage well integrity. Restricting pore network interconnectivity and flaw size/density directly preserves the cement's barrier functionality against hydrocarbon or gases leakages under high pressure differentials for prolonged periods. Moreover, it helps to enhance the stability of the wellbore which will reduce the likelihood of collapse or deformation.

4. Conclusion

This pioneering work evaluated fly ash and eggshell powder as an additive in ilmenite-weighted oil well cement slurries. The particle size distributions and elemental compositions of the fly ash, eggshell powder, ilmenite, and class G cement powders were characterized via XRF analysis. The particles demonstrated potential to enhance cement slurry properties. Moreover, Utilization of fly ash and eggshell's waste products as a sustainable underground gas storage or oil and gas wells material

can solve many problems related to disposal. Also, these waste products are affordable, economical additive for field applications. Based on the results, 2% FA/ESP (75% + 25%)- S2 by weight of cement (BWOC) is optimal. At this concentration:

1. The elastic properties were improved with addition of fly ash and eggshell powder, as evidenced by reducing Young's modulus by 32.24% (from 21.4 to 12 GPa) and Poisson's ratio increases by 35.7% (0.28 to 0.38 GPa). This shows that the combination of fly ash and eggshell powder imparts greater durability and resilience to deformation and stress.
2. The incorporation of fly ash and eggshell declined porosity and permeability of the cement matrix values by 10.7% (from 26.3% to 15.6%) and 58.62% (from 0.029 to 0.012 mD), respectively, preventing mitigating fluid migration issues and secondary cementing risks.
3. The addition of fly ash and eggshell powder to well cement slurry formulation improved compressive strength by 32.13% (from 44.4 to 58.67 MPa).
4. Gel strength and Yield stress increased by 145.7% (from 7.0 to 17.2 lb./100 ft²) and 77.82% (from 23.9 to 42.5 lb./100 ft²), respectively, while plastic viscosity decreased by 1.5% (from 254 to 250.2 cP)
5. Conclusively the combination of fly ash and eggshell powder concentration yields superior mechanical integrity, lower permeability, and improved elastic behavior, demonstrating excellent potential for advancing well cement performance and longevity.

CRediT authorship contribution statement

Rockson Aluah: Writing – original draft, Visualization, Validation, Methodology, Investigation. **Opeyemi Oni:** Validation, Methodology, Investigation, Data curation. **Adesina Fadairo:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Data curation, Conceptualization. **Prasad Pothana:** Investigation.

Declaration of competing interest

There is no conflict of interest between the authors.

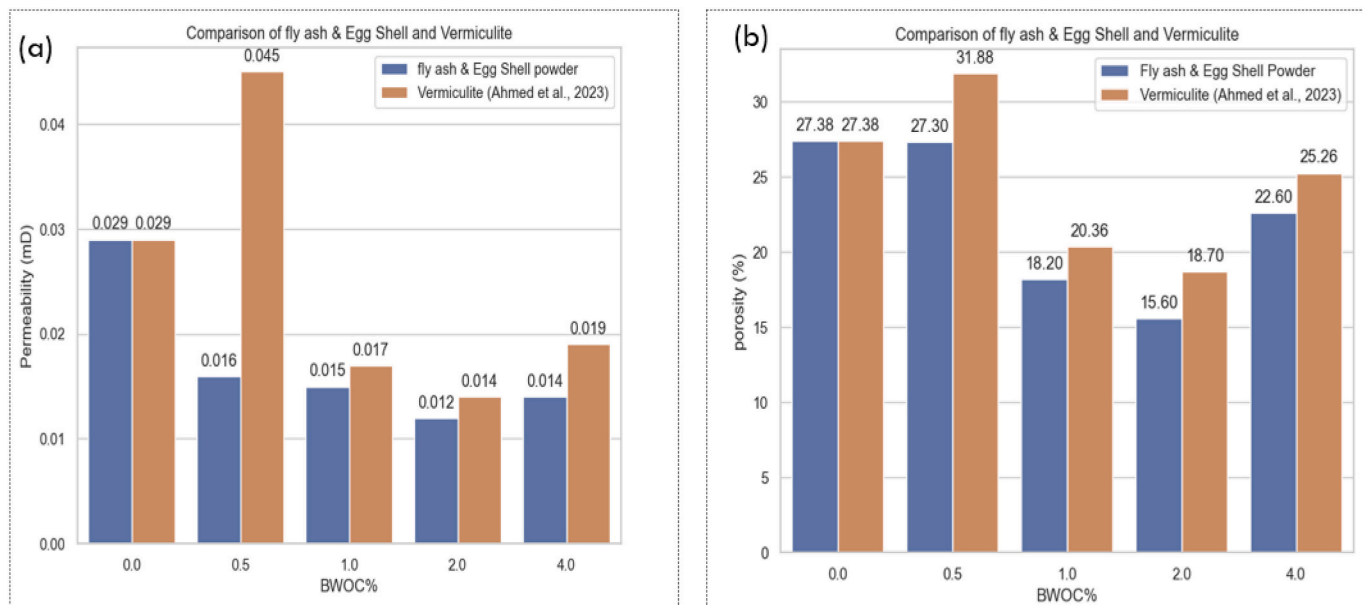


Fig. 21. Effect of final formulations of fly ash and eggshell powder concentration on Petrophysical properties; (a) Porosity; (b) Permeability.

Data availability

No data was used for the research described in the article.

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Further reading

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